WAGER: An Exact Decomposition of Model-Pair Score Gains on Benchmarks with Strong Label Priors

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Abstract

A scene graph generation method reports a ten-point gain in mean recall; a long-tailed classifier reports two points of accuracy. What did the gain buy? On benchmarks whose labels are partly determined by a feature both compared models observe — the object-class pair in relation prediction, the class frequency in long-tailed recognition — such a gain mixes two improvements: the new model may fit the label frequencies *within* groups of that feature more closely, which a frequency table also achieves and a shift in those frequencies removes, or it may match its predictions to the individual cases they were made for. Mean recall, zero-shot recall and significance tests all leave the two mixed. We decompose the gain exactly. Score each model's prediction for one test case against the label of a *different* case in the same group: averaging over such relabellings gives the *transported gain*, and what remains is a *within-group covariance gain*. The identity is exact in finite samples, needs no fitted baseline, no held-out fold and no tuning parameter, and costs $O(NK)$, so any pair of frozen models can be audited from cached predictions. The remainder is an order-two U-statistic, holds for every Bregman score, and admits cluster-robust interval estimation. Applied to two released MOTIFS checkpoints on VG150 with the original authors' code, a ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly transported, with a covariance remainder an order of magnitude smaller under either of two proper scores. The unmodified estimator transfers to

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a frozen-CLIP variant and to long-tailed image and text classification, exposing compositions that aggregate scores hide — including a model that loses on every aggregate measure while discriminating individual cases significantly better than the model it appears to lose to.
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1. Introduction

A scene graph generation method reports a ten-point improvement in mean recall on Visual Genome. A long-tailed classifier reports two points of accuracy on CIFAR-100-LT. What did those gains buy?

The question is not rhetorical, because on benchmarks like these a model can raise its score without becoming any better at the cases it is scored on. In relation prediction, the ordered object-class pair already predicts the relation strongly: given (man, surfboard), the label is very often riding, and a lookup table of training frequencies is famously competitive [1]. A new model can therefore improve by estimating $\Pr(Y \mid \text{class pair})$ more accurately — a real improvement, but one that no more distinguishes *these two men on surfboards* than the frequency table did, and one that evaporates the moment the relation frequencies shift. Or it can improve by telling those two cases apart: putting higher probability on the correct label for the individual image, beyond what the class pair alone implies. The reported score is the sum of the two, and the field's own metrics cannot separate them — not mean recall, not zero-shot recall, and not a significance test on the difference, because both components sit inside the same number.

The problem is general and the community knows it. Frequency baselines are competitive on Visual Genome [1]; VQA systems collapse under changed answer priors [2, 3]; standard metrics obscure tail behaviour [4]; and benchmark audits keep finding multimodal models exploiting non-visual shortcuts that inflate exactly this kind of aggregate score [5, 6]. The reporting standards have responded — mean recall alongside recall, broken down by predicate frequency [7, 4], zero-shot recall on unseen triplets [8], GQA-OOD stratified by answer rarity [9] — and every one of those is a prior-aware slicing of a *single* model's performance. None attributes the gain between two submitted systems.

The customary audit builds a prior-only reference: fit $\widehat{\Pr}(Y \mid \phi)$ from training data, score it, and see how far the full models beat it [1]. As a diagnostic for one model that is useful. As an answer to the pairwise question it is indirect and noisy — the reference has to be estimated, it needs a smoothing rule for rare cells and a model family that may match neither compared system, and decomposing two models separately then subtracting gives no uncertainty for the difference, which is the quantity actually in dispute.

What this paper does.. We decompose the score difference itself, exactly, with no fitted reference at all. The idea takes one sentence. Both models have already emitted a full probability vector for every test case, so each model's score can be recomputed against *any* label, not only the one that was observed. Take two test cases in the same group — sharing the same value of a declared grouping variable ϕ, such as the object-class pair — and score each case's predictions against the *other* case's label. That relabelling preserves the group and its label frequencies while destroying the correspondence between a prediction and the case it was made for. Whatever score advantage survives is an advantage the new model holds over the group as a whole; whatever disappears was case-specific. Averaging within groups turns this into an identity,

$$\text{total score gain} = \underbrace{\text{transported gain}}_{\text{survives relabelling}} + \underbrace{\text{within-group covariance gain}}_{\text{destroyed by relabelling}},$$

exact in the observed sample, for any proper score, with no approximation, no held-out fold and no tuning parameter, at a cost of $O(NK)$ for N cases and K labels. Figure 1 shows the relabelling on two real Visual Genome cases, and Section 2 works the arithmetic through on four data points — including, in Section 2.1, a model-selection decision the split gets right and the aggregate score gets wrong.

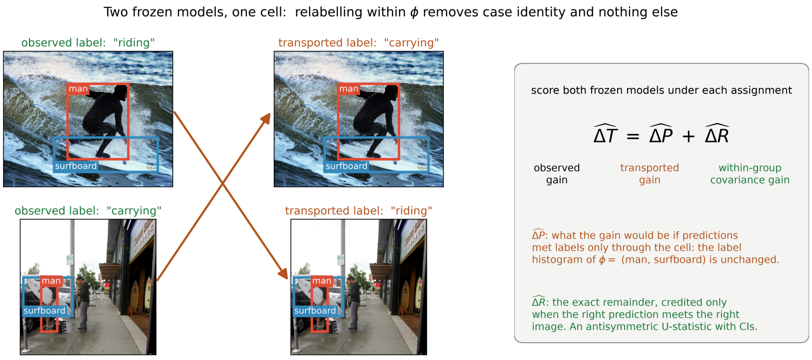

Figure 1: **The construction, on two real VG150 test relations.** Both share the group $\phi = (\mathsf{man}, \mathsf{surfboard})$ — the ordered object-class pair that already predicts the relation — yet carry different labels. Exchanging their labels (arrows) leaves the group and its label frequencies intact while breaking the link between each prediction and the image it was made for. Scoring two frozen models before and after the exchange splits their score difference exactly into a transported gain and a within-group covariance gain. Nothing about the construction is specific to relation prediction: it needs the two probability vectors, the labels, and the group identifiers.

What we find.. Pointed at two publicly released scene-graph checkpoints rather than at predictors of our own, the decomposition gives a precise account of a widely cited debiasing method. Once the two models are matched for confidence, essentially the entire proper-score difference between MOTIFS-TDE and the MOTIFS baseline it debiases is transported gain, with a within-group covariance remainder at least an order of magnitude smaller under either of two proper scores (Section 5.2). A metric the field reads as evidence about rare predicates moves ten points while the quantity that would measure case-level evidence barely moves at all. Since TDE's own construction subtracts a context-only prediction, this characterises the method rather than indicting it — but it does mean the headline number is not the evidence it is generally taken for.

What the second component is.. The component destroyed by relabelling is not a new invention, and it is worth naming correctly because the name predicts how it behaves. Written out, it equals the covariance, inside each group, between how much the new model moved its predicted probability for a label and whether that label actually occurred (Theorem 1). A covariance between a forecast and an outcome indicator is the object isolated by the covariance rearrangement of a proper score that goes back to Yates [10], so we call our second component the *within-group covariance gain*.

It is close to, but not the same as, the *resolution* term of the classical reliability–resolution partition of a proper score [11, 12, 13], and the difference has a practical consequence. Classical resolution depends on a forecast only through the partition its level sets generate, so it is invariant under any strictly monotone relabelling of the forecast values. A covariance between forecast *values* is not: rescaling a fixed model's confidence changes it while changing no ranking, no top-1 decision and no information. The two coincide exactly when both models are calibrated within groups. Section 3.1 states the discrepancy and Section 5.1 measures it — large enough that every comparison we report is between confidence-matched models. Models that differ sharply in calibration, which released checkpoints routinely do, cannot be audited on their raw outputs.

We call the estimator WAGER; the acronym is a label, not a claim, and we do not expand it. Its two components are the *transported gain* ΔP — named for the operation that produces it, not for an interpretation of it — and the within-group covariance gain ΔR.

1.1. Contributions

1. **An audit of a released debiasing method.** Applied to the two publicly released MOTIFS checkpoints of Tang et al. [7] on the canonical VG150 PredCls split, evaluated with the original authors' code, the decomposition shows that TDE's ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly transported, with a within-group covariance remainder at most a small fraction of it (Section 5.2). This is a statement about somebody else's published method, on the benchmark the field actually uses, and it is the paper's headline result.
2. **A drop-in tool for any closed-set benchmark.** The estimator needs four aligned arrays — two $N \times K$ probability matrices, the labels, the group identifiers — runs in $O(NK)$ on cached predictions with no model refitting, no held-out fold and no tuning parameter, and returns both components with cluster-robust intervals in one pass. We apply it unchanged to relation

prediction, to a frozen-CLIP variant of it, to long-tailed image classification across seeds and imbalance ratios, and to long-tailed text classification; only the declared ϕ changes. Code, cached model outputs and a script that verifies every reported number are released.

3. **An exact identity, and what its remainder is.** For any score the observed gain splits exactly — in the population (Theorem 1) and in the finite sample (Theorem 3) — into a transported gain and a remainder that is a U-statistic of order two. For every Bregman score, so for the quadratic and logarithmic scores as named cases, that remainder is a within-group covariance between the models' predicted-probability difference and the label indicator (Theorem 2). A change that is constant within groups contributes exactly zero. The transported component splits in turn into a mean-forecast-fit term minus a dispersion term (Proposition 1), which is why we do not call it a measure of prior fit.
4. **Inference, and the design choices it settles.** We derive the influence function of the difference, giving image-cluster-robust intervals, plus a group-stratified randomization test; the estimator is asymptotically normal with a consistent sandwich variance (Theorem 4), and at a trivial ϕ it reduces to a clustered Diebold–Mariano test of the paired score differential (Corollary 3). Three exact results remove guesswork: using a group's own label frequency as the counterfactual attenuates the estimate by exactly $(n_c - 1)/n_c$, which is why no held-out fold is needed (Proposition 4); merging groups shifts the estimand by an explicit between-group covariance of either sign, so a coarser ϕ is not harmless (Proposition 2); and a Cauchy–Schwarz bound limits how far an unrecorded grouping variable could move the result (Proposition 3). Proofs are in the supplementary material.
5. **Validation, and honest limits.** Simulations establish interval coverage under image clustering and delimit what the two components do *not* separate: prior-only improvements stay out of the covariance channel, unrecorded within-group shortcuts enter it, and monotone recalibration moves score

Table 1: The paper's vocabulary, and what it corresponds to in standard terms.

Term used here	Standard counterpart	Meaning
grouping variable ϕ	stratifying / conditioning variable	a discrete function of the input, declared before the audit, whose levels the labels depend on
cell c	level of ϕ; stratum	the set of test cases sharing one value of ϕ
label transport	within-stratum relabelling; permutation within blocks	replacing a case's label with another case's label from the same cell
total gain ΔT	paired score differential	the observed mean score of the new model minus that of the old
transported gain ΔP	no exact counterpart; a mean-forecast-fit term minus a dispersion term (Proposition 1), *not* a measure of prior fit	the part of ΔT that survives label transport
within-group covariance gain ΔR	the covariance term of a covariance rearrangement of a proper score [10]; twice a classical resolution difference *only* under within-cell calibration	the part destroyed by transport; a within-cell prediction–label covariance
coverage	identified fraction	share of test cases in cells of size ≥ 2, the only cells in which ΔR is identified
calibration-matched	temperature-scaled on held-out data	both models rescaled to comparable confidence before the audit

between the channels — which is why confidence matching is part of the procedure and not a caveat. We also show that equally fine, fully identified partitions can disagree without limit, so the choice of ϕ is a real modelling decision rather than a formality.

1.2. Terminology

The construction needs few new words, and Table 1 fixes them against their standard statistical counterparts. Two of the rows carry a caveat rather than a synonym, because no standard term matches exactly: ΔP is named for the operation that produces it, and ΔR is a covariance rearrangement's term rather than the classical resolution it is often mistaken for. We use "group" and "cell" interchangeably — "cell" where the contingency-table sense is natural, "group" where a reader outside that tradition would prefer it.

1.3. Scope, and what is not claimed

The *identity* needs exactly four aligned arrays: two $N \times K$ matrices of predicted probabilities, the N observed labels, and the N cell identifiers. Any closed-set probabilistic prediction task supplies them. The *procedure* we actually recommend needs more — cluster identifiers for the intervals, and, whenever the compared models differ visibly in confidence, a random split of the clusters and a temperature fitted per model on the held-out half (Section 5.1). We state the distinction here because the four-array framing, taken alone, understates what a faithful audit costs. Free-form or generative evaluation, where no K-way probability vector exists, is out of reach altogether, and we make no claim there.

Four limits on interpretation are worth stating before any result rather than after all of them. First, everything is conditional on the declared ϕ: a positive ΔR says the new model predicts individual cases better *among cases sharing that feature*, and nothing about causality, reasoning, or robustness. Second, ΔR credits any signal that varies within a cell, including a shortcut the analyst did not record. Proposition 3 bounds that exposure rather than removing it, and the bound is not always comfortable: for the Visual Genome pair of Appendix 5.3 an unrecorded covariate with an aggregate correlation of 0.061 would account for the entire reported gain (Section 4.6). Third — and this is the limitation we would most want a reader to carry — ΔR is *not invariant to recalibration*. Rescaling a fixed model's confidence changes it while changing no ranking and adding no information about any case, and the effect is large enough to reverse verdicts: in simulation a pure recalibration moves ΔR by -0.488, tens of times the size of any real gain reported in this paper. Everything we report is therefore for confidence-matched pairs, by the protocol of Section 5.1, and a reader who applies the estimator to unmatched models should expect the split to be dominated by their confidence difference. Fourth, a transported gain is not illegitimate — fitting the deployment label distribution better is real skill whenever that distribution is the one that matters — and it is also not purely a matter of prior fit (Proposition 1). The decomposition says where a gain lives, not whether it is worth having.

Table 2: A four-point example in a single cell. Model A moves every case to the cell's label frequencies; model B moves each case towards its own correct label by the same amount. The two models are equally far from q_0, and only B uses information about the individual case.

case i	label y_i	$q_0(\cdot \mid x_i)$	model A: $q_1(\cdot \mid x_i)$	model B: $q_1(\cdot \mid x_i)$
1	1	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.75, 0.25)$
2	1	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.75, 0.25)$
3	1	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.75, 0.25)$
4	2	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.25, 0.75)$

Organization.. Section 2 works a four-point example and, in Section 2.1, uses it to exhibit a decision the split changes. Section 3 places the construction in the literature on proper-score decomposition and forecast comparison. Section 4 develops the estimator and its theory. Section 5 validates it and audits two released checkpoints. Sections 6 and 7 discuss limits and conclude. Proofs are in Appendix A; three further empirical studies are in Section 5.

2. A worked example

Four data points are enough to show the whole construction, and to exhibit in arithmetic the main properties Section 4 proves in general. Take one cell containing four test cases and two possible labels. Three of the cases carry label 1 and one carries label 2, so the cell's label frequencies are $(3/4, 1/4)$. The old model q_0 is uninformative within the cell: it predicts $(0.5, 0.5)$ for all four cases. We compare two candidate replacements, shown in Table 2.

Score both models with the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, and for each case tabulate the score *difference* at each of the two possible labels, not only at the observed one. Writing $h_i(y) = S(q_1(\cdot \mid x_i), y) - S(q_0(\cdot \mid x_i), y)$, all the arithmetic is in eighths:

$$h_i(y) = \begin{cases} +3/8 & \text{if } y \text{ is the label the new model moved towards,} \\ -5/8 & \text{otherwise.} \end{cases} \tag{1}$$

This vector, evaluable at labels that were never observed for case i, is the only object the method needs beyond the labels themselves.

Model A: a pure prior refit.. Every case gets the same prediction, so $h_i(1) = +3/8$ and $h_i(2) = -5/8$ for all four cases. The *observed* mean gain uses each case's own label,

$$\Delta T = \tfrac{1}{4}\left[3 \cdot \tfrac{3}{8} + 1 \cdot \left(-\tfrac{5}{8}\right)\right] = \tfrac{1}{8}. \tag{2}$$

Now transport labels: replace each case's label, in turn, by each of the other three cases' labels and average. Case 1 is scored against labels $\{1, 1, 2\}$, giving $\frac{1}{3}(\frac{3}{8} + \frac{3}{8} - \frac{5}{8}) = \frac{1}{24}$, and likewise for cases 2 and 3; case 4 is scored against $\{1, 1, 1\}$, giving $+\frac{3}{8}$. The mean transported gain is

$$\Delta P = \tfrac{1}{4}\left[3 \cdot \tfrac{1}{24} + \tfrac{3}{8}\right] = \tfrac{1}{8}, \qquad \text{so} \qquad \Delta R = \Delta T - \Delta P = 0. \tag{3}$$

Model A's entire advantage survives the relabelling, and its covariance gain is exactly zero — not approximately, and not zero on average over hypothetical resamples, but zero in this sample. That is the general fact: a prediction change that is constant within a cell contributes nothing to ΔR (Theorem 1). Note also that $\Delta P = 1/8$ is the largest transported gain any within-cell-constant prediction can earn here, because a proper score is maximized in expectation at the true frequencies $(3/4, 1/4)$ — which is exactly what model A predicts.

Model B: a case-specific improvement.. Cases 1–3 are unchanged, but case 4 now moves towards label 2, so $h_4(1) = -5/8$ and $h_4(2) = +3/8$. Every case is now scored at $+3/8$ on its own label, so $\Delta T = 3/8$. Under transport, cases 1–3 are unaffected ($\frac{1}{24}$ each), but case 4 is scored against three label-1 cases at $-5/8$ apiece. The mean transported gain turns negative,

$$\Delta P = \tfrac{1}{4}\left[3 \cdot \tfrac{1}{24} - \tfrac{5}{8}\right] = -\tfrac{1}{8}, \qquad \Delta R = \tfrac{3}{8} - \left(-\tfrac{1}{8}\right) = \tfrac{1}{2}. \tag{4}$$

Two things are worth reading off this. First, model B's aggregate gain of $3/8$ *understates* its case-level improvement of $1/2$, and the reason is *not* that B fits the cell's label frequencies worse. It fits them better: averaged over the cell it predicts 0.625 for label 1 against the old model's 0.5, and the truth is 0.75, so its mean-forecast fit improves by exactly 0.09375. What offsets that is dispersion. B's forecasts vary inside the cell where the old model's did not, and by Proposition 1 the transported channel charges that variation at exactly the same 0.09375; the two brackets cancel, so the transported gain computed with the cell's own frequencies is exactly zero. The $-1/8$ the estimator reports is what remains after each case is scored against the *other* cases' labels only, which removes the $\frac{1}{n_c}\widehat{\Delta R} = 1/8$ that the in-sample plug-in had credited to the transported channel. Both numbers are right; they answer slightly different questions, and Proposition 4 relates them exactly.

Second, an evaluator reading only the total would credit model B with three times model A's improvement, whereas the decomposition shows the two improving on disjoint grounds: A earns $1/8$ purely by moving to the cell's frequencies and gains no covariance at all, while B's $1/2$ of covariance gain is partly concealed by a transported term that turns negative. This is the failure mode that motivates the paper, in miniature: aggregate scores are not merely imprecise about composition, they can order two models by the wrong criterion.

The covariance form, and why leaving one out matters.. Theorem 1 says ΔR is a within-cell covariance between the probability change and the label indicator. Computing that covariance directly for model B, over the four cases, gives $2\sum_y \mathrm{Cov}(\Delta q_y, \mathbb{1}\{Y = y\}) = 3/8$ — not the $1/2$ the transport calculation returned. The discrepancy is not an error in either: the covariance uses the cell's *own* four labels to form the reference frequencies, so each case contributes to the reference it is compared against. That in-sample plug-in is attenuated by exactly $(n_c-1)/n_c$, here $3/4$, and indeed $\frac{3}{4}\cdot\frac{1}{2} = \frac{3}{8}$. Transport, which scores each case only against the *other* cases' labels, removes the attenuation without any held-out data. Proposition 4 shows this holds exactly, for every cell with $n_c \geq 2$ and every grouping variable, which is why the method needs no sample splitting. The example is reproduced as a unit test in the accompanying software.

One property the example cannot show, because both of its candidate models happen to be as confident as each other, is the estimator's sensitivity to confidence itself. Multiply every prediction's odds by a constant and ΔR moves, even though no case has been reordered and no information added. That is a real limitation rather than a technicality — Section 5.1 measures it at -0.488 in a design where the two models carry identical case-level information — and it is why every result in this paper is reported for confidence-matched pairs.

2.1. A decision the split changes

A decomposition that no decision depends on would be an ornament. The four points above are enough to show one, and to show that the split predicts it where the aggregate score cannot.

Read at the benchmark's own label frequencies, both candidate models improve on q_0 and model B improves by three times as much: $\Delta T = 1/8$ against $3/8$. An evaluator choosing between them on the reported score picks B, and is right to. Now suppose the system will be deployed where the label frequencies in this cell are $(\frac{1}{4}, \frac{3}{4})$ rather than the $(\frac{3}{4}, \frac{1}{4})$ the test set happens to exhibit — a label shift, the situation post-hoc prior correction exists to address [14, 15]. Reweighting each case by $p'(y_i)/p(y_i)$ and rescoring gives

$$\text{model A}: \; +\tfrac{1}{8} \;\longrightarrow\; -\tfrac{3}{8}, \qquad \text{model B}: \; +\tfrac{3}{8} \;\longrightarrow\; +\tfrac{3}{8}. \tag{5}$$

Model A does not merely lose its margin; it becomes *worse* than the model it was meant to replace, while model B is untouched. The ranking of A against q_0 reverses on a change in the deployment distribution that leaves every model and every prediction exactly as it was.

The decomposition says in advance which of the two is exposed, and the aggregate score does not. A's entire gain sits in the transported channel, which is the channel a change in label frequencies acts on; B's sits in the covariance channel, which such a change leaves alone, because scoring each case against its own label is unaffected by how often that label occurs once the case is reweighted. This is the operational content of the split: $\widehat{\Delta P}$ is the part of a reported improvement that is contingent on the evaluation set's label frequencies matching the deployment's, and $\widehat{\Delta R}$ is the part that is not. Two systems with the same headline gain and different compositions are different bets, and only the decomposition prices them differently. Section 6 returns to what this does and does not license, and Appendix 5.6 exhibits the same structure on real models, where correcting a model's within-cell prior post hoc registers as almost pure transported gain (+0.02401 against +0.00111) exactly as a correction carrying no case-level information must.

3. Related work

The construction sits between two literatures that rarely meet: the decomposition of proper scores, which supplies the object being estimated, and the comparison of forecasters, which supplies the inferential target. We take those first, then the applied literature that motivates the problem.

3.1. Relation to classical proper-score decompositions

Conditional on a chosen grouping variable, a proper score splits into a calibration (reliability) term and a resolution term [11, 12], made conditioning-set explicit by Bröcker [13]. Murphy [11] is already the vector, multicategory partition and Bröcker [13] already the general strictly-proper one, so the extensions to multiclass outcomes and general scores that recent work re-derives [16] are refinements of those results rather than the first statements of them; the partition also continues to be extended in other directions, to divergence-based uncertainty quantification [17] for instance.

Our ΔR is related to that resolution term but is not equal to it, and the distinction matters enough to state before anything is built on it. The representation of a resolution-like quantity as a *covariance* between the forecast and the outcome indicator is due to Yates [10], whose covariance decomposition of the Brier score is the direct ancestor of Eq. (12); a recent rearrangement revisits it [18]. The classical resolution term of DeGroot and Fienberg [12] is a different object: it depends on a forecast only through the partition generated by its level sets, and is therefore invariant under any strictly monotone relabelling of forecast values.

A covariance between forecast values and outcomes has no such invariance. The two coincide — ΔR_c equals twice a difference of classical resolution terms — exactly when both models are calibrated within each cell of ϕ, and differ otherwise by a reliability-resolution cross term that a monotone recalibration moves. Section 5.1 measures the gap directly: a pure recalibration of one model, which cannot change either model's classical resolution at all, moves $\widehat{\Delta R}$ by -0.488 in the simulation design and reverses the sign verdict of the audit in Section 5.2. We therefore name ΔR for the covariance it is, treat the classical resolution reading as the special case it is, and adopt calibration matching as procedure rather than caveat.

What the classical decomposition does not supply is inference for a pairwise contrast. The point estimates do combine: both models share the same ϕ-partition and hence the same per-cell n_c, so the same $(n_c - 1)/n_c$ attenuation (Proposition 4) applies to each model's separately estimated in-sample resolution, and subtracting two naive plug-ins recovers exactly the biased estimate of the contrast, term for term. We do not claim this is the same correction that Ferro and Fricker [19] apply to a single forecaster's decomposition: theirs is an additive adjustment to a differently parameterized resolution estimate and ours a multiplicative factor on a leave-one-out contrast, and we have not established an equivalence between them. What we do claim is narrower and checkable, namely that Proposition 4 gives the exact factor for the contrast, which is what the estimator needs. Inference, however, does not transfer. Variance estimators for the single-model decomposition [20] target one resolution term in isolation and supply no covariance between it and a second model's covariance estimated from the same data, so subtracting two single-model estimates leaves the contrast's own sampling variability unaddressed; a variance for the difference would have to be rebuilt from the joint law of the two marginal estimators. That rebuilding is a routine exercise rather than an open problem, and the field is not silent about it: Ferro [21] compares probabilistic forecasting systems through the Brier score directly, DelSole and Tippett [22] treat the comparison of forecast skill as an inference problem in its own right, Siegert et al. [23] give tests and a power analysis for detecting skill improvements, and the paired-U-statistic covariance construction of DeLong et al. [24] is exactly the joint-law argument in another setting. What is new here is not that a contrast can be given a variance, but that the contrast admits an *exact finite-sample decomposition* whose remainder is an antisymmetric U-statistic with a one-pass influence function — so the split and its uncertainty come from one construction rather than from combining two marginal ones. Forming the paired gain vector first and targeting the contrast directly makes the influence function that of the *difference* itself, in one pass, and is also what makes the exact finite-sample identity and the antisymmetric kernel available without constructing two single-model estimators at all.

3.2. Comparing two forecasters

Inference on a paired score differential is the classical territory of comparative forecast evaluation: the Diebold–Mariano test [25] and conditional predictive ability tests [26] ask whether one forecaster beats another, optionally conditional on covariates. That is the closest existing relative of a gain evaluated conditional on ϕ, and Corollary 3 makes the relation exact: at the trivial one-cell partition our total-gain statistic and its interval *are* a clustered Diebold–Mariano test, corrected for cluster rather than serial dependence. The decomposition is therefore best read as everything those tests already deliver plus a split that neither provides. A structurally close precedent for comparing two correlated models through a paired U-statistic exists outside forecast evaluation as well: DeLong et al. [24] compare two correlated ROC curves' areas via a generalized U-statistic covariance estimator, the same construction applied to an unconditional ranking statistic rather than a declared-cell gain.

Two further literatures locate the same object from other directions. The transported component is precisely what label-shift procedures manipulate post hoc: re-weighting a classifier's outputs towards a new label distribution [14, 15] can add or remove transported gain without touching case-level evidence, which Appendix 5.6 exploits as a falsification test. And evaluating calibration within cells of a declared grouping, uniformly over many candidate groupings, is the subject of multicalibration [27], which is the formal treatment of the multiplicity question raised by the choice of ϕ (Section 6).

3.3. Transport, permutation, and U-statistics

Strictly proper scoring rules reward honest predictive distributions [28]; a recent survey revisits estimation and forecast evaluation under them broadly [29]. Permutation methods have long been used to assess classifier performance and conditional independence [30, 31], with recent work characterizing when generalized permutation tests attain optimal power [32] and extending conditional-independence testing through ball divergences [33] and transport-map reductions to the unconditional case [34]. Conditional permutation tests generally estimate or assume a conditional distribution when the conditioning variables are continuous or high-dimensional. The setting here is different in a way that buys exactness: the benchmark supplies a *discrete* grouping variable, and the target is not generic conditional independence but one specific model-to-model score gain, so within-cell transport requires no nuisance law at all. The remainder is a U-statistic of order two in the sense of Hoeffding [35], and its Hájek projection places the inference within the same body of theory that recent work extends to anytime-valid confidence sequences [36]. Pairwise betting has also been used to test exchangeability sequentially [37]; ours is a fixed-sample decomposition and makes no sequential-validity claim. The new element is the gain-contrast transport identity and the attribution it licenses, not permutation or U-statistic theory by itself.

A more classical connection concerns what an unrecorded grouping variable can hide. Omitted-variable sensitivity analysis in observational causal inference [38] and its correlation- and moment-parameterized refinements [39, 40] bound how far an unmeasured confounder could move a causal estimate; Proposition 3 adapts the same Cauchy–Schwarz argument to bound how far an unrecorded covariate could move ΔR — the same question asked of a different estimand.

Usable-information analyses measure information accessible to a chosen predictor family [41, 42, 43]. Their central objects are model- or family-level quantities; ours is deliberately narrower and more operational, attributing the score difference between two concrete frozen models. We do not identify conditional mutual information, and a positive ΔR establishes nothing about causal or compositional structure.

3.4. Where the problem arises

Benchmarks with strong label priors are the applied motivation, and the applications in Section 5 and Section 5 come from computer vision and natural-language classification. Dataset bias has long complicated comparisons between recognition systems [44, 45]. In scene graph generation a frequency predictor of $p(\text{predicate} \mid \text{subject}, \text{object})$ is surprisingly competitive [1]; mean recall and causal interventions were proposed to reduce that dependence [46, 47, 7], though metric pathologies remain [4] and the same diagnosis recurs in open-vocabulary and knowledge-enhanced work [48, 49, 50]. In visual question answering, GQA-OOD stratifies evaluation by answer rarity within question groups [9] and VQA-CP changes language priors between train and test [3], though changing-prior splits can themselves be exploited [51]; the same shortcut is still targeted directly [52, 53], and benchmark-level audits of exploitable non-visual shortcuts have widened to multimodal evaluation generally [5, 6]. In long-tailed recognition a classifier can raise aggregate accuracy by fitting the training class frequencies more precisely without discriminating better within a class [54, 55]; the canonical remedies rebalance the objective or the classifier [56, 57, 58, 59], with a continuing stream of variants [60, 61].

The reporting standards in these fields have themselves responded to the shortcut: mean recall is reported alongside recall and often broken down by predicate frequency [7, 4], and zero-shot recall [8] checks whether a gain survives on triplets absent from training. These are all effective prior-aware slicings, splits, or stress tests of a *single* model's performance. What none provides is what we target: an exact decomposition, with uncertainty, of the score difference between two fixed systems. A stratified or zero-shot metric can show *that* a gain is uneven across the prior; it does not attribute the pairwise gain itself, on the full evaluation set, with a finite-sample identity and inference. The remedies, equally, intervene on how a model is trained rather than on how a trained pair is compared.

4. The estimator and its properties

This section states the construction of Sections 1–2 formally and establishes its properties. The order is: notation and the object being decomposed (Section 4.1); the population identity and the covariance form of its remainder (Section 4.2), which Section 4.4 shows is not specific to the quadratic score; what happens when the grouping variable is made coarser (Section 4.5) or is missing a relevant covariate altogether (Section 4.6); the finite-sample identity and its $O(NK)$ computation (Sections 4.7–4.8); why the counterfactual leaves each case out rather than fitting the group's own frequencies (Section 4.9); and inference, both finite-sample and asymptotic (Sections 4.10–4.11). Proofs are in Appendix A. A reader who wants only what is needed to use the estimator can read Sections 4.1, 4.2, 4.7 and 4.10 and skip the rest.

4.1. Setup: what is being decomposed

Let $q_0(\cdot \mid x)$ and $q_1(\cdot \mid x)$ be the predictive distributions of an old and a new frozen model over K labels. Let $S(q, y)$ be a bounded proper score, oriented so that larger is better, and let $C = \phi(X)$ be a declared discrete grouping variable. In the relation-prediction example of Section 1, C is the ordered subject–object class pair; in a long-tailed classification benchmark it is a coarse superclass. We call a value of C a *cell* and require it to be declared before the audit is run. Define the instance-specific *gain vector*

$$H_x(y) = S(q_1(\cdot \mid x), y) - S(q_0(\cdot \mid x), y), \qquad y = 1, \ldots, K. \tag{6}$$

Unlike the realized gain $H_X(Y)$, this vector can be evaluated at a counterfactual label without rerunning either model.

Our default is the quadratic score

$$S_{\mathrm{B}}(q,y) = 2q_y - \|q\|_2^2, \tag{7}$$

which differs from negative Brier loss only by a label-only constant. It is strictly proper and bounded, avoids probability clipping, and uses the entire predictive distribution. A floored log score is a sensitivity analysis.

4.2. *Within-cell counterfactual transport*

For a cell c, define three population quantities:

$$\Delta T_c = \mathbb{E}[H_X(Y) \mid C = c], \tag{8}$$

$$\Delta P_c = \mathbb{E}_{H|c}\,\mathbb{E}_{Y'|c}[H_X(Y')], \tag{9}$$

$$\Delta R_c = \Delta T_c - \Delta P_c. \tag{10}$$

Here Y' is an independent label drawn from the same cell. Thus ΔT_c is the observed score gain, ΔP_c is the gain retained after breaking the instance-specific prediction–label assignment while preserving the cell, and ΔR_c is the covariance gain lost by that transport.

Theorem 1 (Transport and covariance identities)**.** *For any integrable score contrast H,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \qquad \Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{11}$$

For the quadratic score, writing $\Delta q_y(X) = q_1(y \mid X) - q_0(y \mid X)$,

$$\Delta R_c = 2\sum_{y=1}^{K} \mathrm{Cov}\Big(\Delta q_y(X), \mathbb{1}\{Y = y\} \mid C = c\Big)\,. \tag{12}$$

Consequently, if the gain vector is a function of c alone, then $\Delta R_c = 0$.

Equation (12) is the operational meaning of WAGER: the new model receives covariance credit only to the extent that its probability change aligns more strongly with the label of the correct instance within the same cell. Cell-constant score improvements, including improved fitting of the cell's label histogram, remain in ΔP. The result does not assert that a nonzero covariance reflects causal or compositional understanding; unrecorded within-cell shortcuts can also contribute, so ϕ must be declared and stress-tested.

4.3. What the transported gain measures

The component that survives transport is easy to misname. It is tempting to call ΔP a prior-fit gain, since transport replaces a case's label by an independent draw from its cell; several earlier drafts of this paper did exactly that. The following identity shows the name is wrong, and says what the quantity is instead.

Proposition 1 (Composition of the transported gain)**.** *Under the quadratic score, write* $\bar{q}_m(c) = \mathbb{E}[q_m(\cdot \mid X) \mid C = c]$ *for model m's mean forecast in cell c,* $\mathrm{Var}(q_m \mid c)$ *for the within-cell covariance matrix of its forecast vector, and* $p(c)$ *for the cell's transport law. Then*

$$\Delta P_c = \underbrace{\left[\|p(c) - \bar{q}_0(c)\|^2 - \|p(c) - \bar{q}_1(c)\|^2\right]}_{\textit{improvement in mean-forecast fit}} - \underbrace{\left[\mathrm{tr}\,\mathrm{Var}(q_1 \mid c) - \mathrm{tr}\,\mathrm{Var}(q_0 \mid c)\right]}_{\textit{increase in within-cell dispersion}}. \tag{13}$$

Only the first bracket is prior fit: it is positive exactly when the new model's average within-cell forecast sits closer to the cell's label frequencies. The second bracket has nothing to do with the prior at all. It rewards the new model for making *less* dispersed predictions inside the cell, and it does so at a fixed mean forecast, so ΔP can be large and positive when neither model's prior fit has changed by anything. A single cell suffices to see it: with two labels and cell frequencies $(\frac{1}{2}, \frac{1}{2})$, let q_0 alternate between $(1, 0)$ and $(0, 1)$ in a pattern uncorrelated with the label, and let q_1 predict $(\frac{1}{2}, \frac{1}{2})$ throughout. Both models' mean forecasts equal the cell frequencies exactly, so the first bracket is zero; the whole observed gain of $\frac{1}{2}$ is transported, produced entirely by the collapse in dispersion. The channel is therefore the reliability-and-sharpness channel of the classical partition rather than a prior-fitting channel, which is why we name it for the operation that produces it and not for an interpretation of it.

Two consequences are worth stating. First, a claim that some gain is "mostly prior-fitting" does not follow from a large $\widehat{\Delta P}$ alone; where this paper reports such a reading it is the two brackets of Eq. (13) that carry it, and Section 5.2 reports them separately for the audited pair. Second, Eq. (13) explains why calibration matching moves $\widehat{\Delta P}$ so sharply (Section 5.1): a temperature change is almost purely a change in dispersion.

4.4. Generalization: the Bregman-score family

Nothing in the proof of Theorem 1 (Appendix A) uses properness: the identity $\Delta T_c = \Delta P_c + \Delta R_c$ and the covariance sum in Eq. (11) follow from the definitions of $\Delta T_c, \Delta P_c, \Delta R_c$ alone, for any score S. What is specific to the quadratic score is the further reduction to Eq. (12): the part of $H_x(y)$ that does not depend on y integrates to zero in the covariance sum because $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The same cancellation occurs for every strictly proper score generated by a Bregman divergence, so Eq. (12) and the log-score identity used as a robustness check in Section 5.2 are two named instances of one theorem rather than an unexplained extra case.

For a strictly convex, differentiable $\psi : \Delta_K \to \mathbb{R}$ on the probability simplex, the associated *Bregman score* is

$$S_\psi(q, y) = \psi(q) + \langle \nabla\psi(q),\, e_y - q \rangle, \tag{14}$$

where e_y is the y-th standard basis vector. For the true label distribution p,

$$\mathbb{E}_p[S_\psi(p, Y)] - \mathbb{E}_p[S_\psi(q, Y)] = \psi(p) - \psi(q) - \langle \nabla\psi(q), p - q \rangle = D_\psi(p, q) \geq 0, \tag{15}$$

the Bregman divergence of ψ between p and q, with equality iff $q = p$ by strict convexity, so S_ψ is strictly proper [28]. Taking $\psi(q) = \|q\|_2^2$ recovers the quadratic score of Eq. (7); taking $\psi(q) = \sum_k q_k \log q_k$ (negative entropy) recovers the log score.

Theorem 2 (Bregman-score covariance identity)**.** *Let* $H_x(y) = S_\psi(q_1(\cdot \mid x), y) - S_\psi(q_0(\cdot \mid x), y)$ *for a Bregman score generated by a strictly convex, differentiable* ψ*, and write* $\Delta g_y(x) = \partial_y \psi(q_1(\cdot \mid x)) - \partial_y \psi(q_0(\cdot \mid x))$ *for the contrast of the y-th gradient coordinate. Then*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}\big(\Delta g_y(X),\ \mathbb{1}\{Y = y\} \mid C = c\big). \tag{16}$$

Corollary 1 (Quadratic and log score)**.** *For* $\psi(q) = \|q\|_2^2$*,* $\Delta g_y(x) = 2\Delta q_y(x)$ *and Eq.* (16) *is Eq.* (12)*. For* $\psi(q) = \sum_k q_k \log q_k$*,* $\Delta g_y(x) = \log q_1(y \mid x) - \log q_0(y \mid x)$*, and*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}\left(\log \frac{q_1(y \mid X)}{q_0(y \mid X)},\ \mathbb{1}\{Y = y\}\ \middle|\ C = c\right). \tag{17}$$

Eq. (17) gives the log-score check of Section 5.2 the same exact covariance meaning that Eq. (12) gives the quadratic score: within-group covariance under the log score is the within-cell covariance between the log-likelihood-ratio contrast of the two models and the observed label indicator, not merely "a sensitivity analysis." Any other Bregman score – the spherical or power-family scores catalogued by Gneiting and Raftery [28], for instance – inherits an identity of the same form through its own ψ, so Theorem 2 fixes the scope of the estimator's scoring-rule dependence once, rather than leaving each alternative score to be checked separately. The implemented log score floors each probability at $\epsilon = 10^{-6}$ to avoid $-\infty$ on an unseen label; Eq. (17) is exact for the unfloored $\log q_y$, and the floored version used throughout is a sensitivity check on that exact identity rather than a literal instance of it, negligibly different whenever no relevant probability is within ϵ of zero.

4.5. Coarsening the cell

Appendix 5.5 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated covariance gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

Proposition 2 (Coarsening decomposition)**.** *Let $\bar{\phi} = g(\phi)$ be any coarsening of the grouping variable, so that each cell $\bar{c}$ of $\bar{\phi}$ is a union of cells of ϕ. Then*

$$\Delta R_{\bar{c}} = \mathbb{E}[\Delta R_C \mid \bar{\phi} = \bar{c}] + \sum_{y=1}^{K} \mathrm{Cov}(\mathbb{E}[H_X(y) \mid C], \Pr(Y = y \mid C) \,\big|\, \bar{\phi} = \bar{c}) . \quad (18)$$

The first term is the coarse cell's share of the fine-grained covariance gain the paper already reports; it is the quantity a reader would expect a coarser audit to recover on average. The second term is new: it is a between-cell covariance, taken over the finer cells nested inside $\bar{c}$, between each fine cell's typical gain at label y and its label-y frequency. It is generally not sign definite, so coarsening ϕ can inflate or deflate the credited covariance gain relative to the finer audit, and Table 7 shows the empirically inflating case. The second term vanishes exactly when the quantities it correlates do not vary across the fine cells nested in $\bar{c}$ – that is, when $\mathbb{E}[H_X(y) \mid C]$ or $\Pr(Y = y \mid C)$ is constant within each coarse cell. Proposition 2 thereby formalizes why the finest defensible grouping variable is the conservative default: relative to the finer audit, coarsening changes the estimate only by adding cross-cell prior structure into ΔR, while the finer partition's covariance gain is preserved on average through the first term. The proof is a per-label application of the law of total covariance, summed over y (Appendix A).

4.6. Sensitivity to an unrecorded confounder

Proposition 2 already describes exactly what happens when a relevant covariate is folded into ϕ: doing so refines the cell. Read in the other direction, the same identity quantifies what happens when a relevant covariate is *never* declared, which is WAGER's main scope limit (Section 6). Let Z be any unrecorded variable and let (ϕ, Z) denote the finer, fully adjusted cell. Applying Proposition 2 with $\bar{\phi} = \phi$ and fine cell $C = (\phi, Z)$ gives, for every declared cell c of ϕ,

$$\Delta R_c(\phi) = \mathbb{E}\Big[\Delta R_{(\phi,Z)} \mid \phi = c\Big] + \sum_{y=1}^{K} \mathrm{Cov}(\mathbb{E}[H_X(y) \mid \phi, Z],\ \Pr(Y = y \mid \phi, Z) \mid \phi = c)\,. \tag{19}$$

The first term is what an analyst who could observe Z would report; the second is the bias of using ϕ alone. Eq. (19) is exact but requires knowing Z's joint law with (H, Y) to evaluate. The following bound turns it into a sensitivity analysis that requires only a judgment about how strongly an unrecorded variable could plausibly act, in the spirit of Rosenbaum [38] and recent partial-correlation sensitivity models for unmeasured confounding [39, 40].

Proposition 3 (Sensitivity bound for an unrecorded confounder)**.** *Suppose $H_x(y) \in [-M, M]$ for all x, y (WAGER already requires a bounded score). Fix a cell c of ϕ, write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$ (both conditional on $\phi = c$) for the two K-vectors on the right of Eq.* (19)*, and let*

$$\rho = \frac{\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)}{\sqrt{\sum_{y=1}^{K} \mathrm{Var}(a_y \mid c)}\sqrt{\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c)}} \in [-1, 1] \tag{20}$$

be their aggregate correlation. Then

$$\begin{aligned}\Big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\Big| \ &\le\ |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(a_y \mid c)}\ \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c)} \\ &\le\ |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(H_X(y) \mid \phi = c)}\ \sqrt{\textstyle\sum_{y=1}^{K} p_y(c)(1 - p_y(c))},\end{aligned} \tag{21}$$

where $p_y(c) = \Pr(Y = y \mid \phi = c)$; the second inequality holds termwise ($\mathrm{Var}(a_y \mid c) \le \mathrm{Var}(H_X(y) \mid c)$ *and* $\mathrm{Var}(b_y \mid c) \le p_y(c)(1 - p_y(c))$ *by the law of total variance, since conditioning on more variables cannot increase the variance of a conditional expectation) followed by monotonicity of the square root applied to each sum separately – a product of two square-root sums, not a sum of per-label square roots, which the law of total variance alone would not license.*

Corollary 2 (Free bound)**.** *Each a_y ranges over $[-M, M]$ as a conditional ex-*

pectation of H, so $\mathrm{Var}(a_y \mid c) \leq M^2$ *and* $\sum_y \mathrm{Var}(a_y \mid c) \leq KM^2$. *The label probabilities satisfy* $\sum_y b_y = 1$*, which gives* $\sum_y b_y^2 \leq 1$ *and, by Cauchy–Schwarz,* $\sum_y (\mathbb{E}b_y)^2 \geq 1/K$*; hence*

$$\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c) = \mathbb{E}\Big[\sum_y b_y^2\Big] - \sum_y (\mathbb{E}b_y)^2 \;\leq\; 1 - \tfrac{1}{K}. \tag{22}$$

Substituting both into Eq. (21) *gives the data-free bound*

$$\Big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\Big| \;\leq\; |\rho|\, M \sqrt{K-1}. \tag{23}$$

Earlier versions of this corollary bounded $\sum_y \mathrm{Var}(b_y \mid c)$ by Popoviciu's inequality applied label by label, giving $K/4$ and hence $|\rho|KM/2$. That discards Eq. (22): the b_y are not K unconstrained quantities in $[0, 1]$ but a point on the simplex, and $K/4$ is unattainable for $K > 2$. At VG150's $K = 50$ and $M = 2$ the corrected free bound is $14.00|\rho|$ rather than $50.00|\rho|$, a factor of 3.57.

An analyst willing to bound $|\rho| \leq \bar{\rho}$ – how strongly an unrecorded variable could jointly move a model's within-cell score gain and the within-cell label distribution – obtains a numeric limit on how far $\widehat{\Delta R}_c$ can be from what a fully adjusted audit would find, without observing or modeling Z, in the spirit of Rosenbaum [38] and recent partial-correlation sensitivity models for unmeasured confounding [62, 39, 40]. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds or risk ratio, it can be elicited the same way across benchmarks.

What the bound says about this paper's own estimates.. Table 3 reports both forms of the bound on the Visual Genome pair of Appendix 5.4, together with the coarsening from the class-pair to the subject-only cell, which instantiates Eq. (19) exactly with Z = object class. The free bound is directionally correct and, at $K = 50$, still 1,887 times the actual coarsening bias, so any $\bar{\rho}$ an analyst would plausibly declare certifies a limit far above what is observed. That looseness is a property of a worst-case inequality applied to label probabilities that concentrate away from the extremes at which it binds, not a defect of this dataset — but it

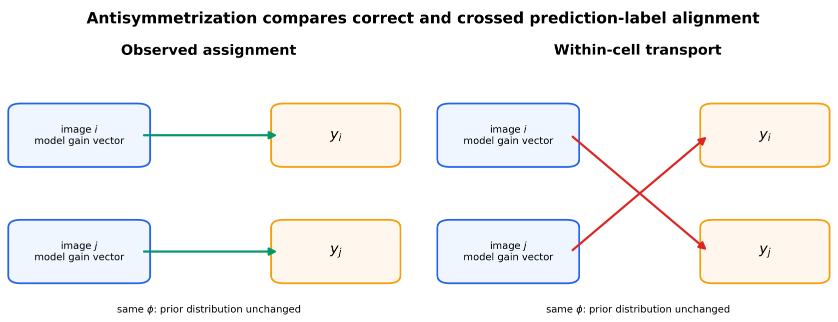

Figure 2: **The antisymmetric pair kernel.** For two examples in the same prior cell, WAGER subtracts the score under crossed labels from the score under the observed assignment. Cell membership and label counts are unchanged; only within-group covariance is removed.

also means the free bound cannot support a claim about whether any particular reported $\widehat{\Delta R}$ would survive confounding.

The sharp per-cell form of Eq. (21), which replaces the worst-case sums by the cell's own $\sum_y \mathrm{Var}(H_X(y) \mid c)$ and $\sum_y p_y(c)(1 - p_y(c))$, can support such a claim, and it is less reassuring than the free bound. Computed by `experiments/sensitivity_analysis.py` from per-instance arrays for the same pair, it gives $B = 0.23284$ at $\bar{\rho} = 1$, so the *robustness value* — the smallest aggregate correlation at which the bound reaches the estimate — is

$$\rho^\dagger = \frac{|\widehat{\Delta R}|}{B} = \frac{0.01427}{0.23284} = 0.06127. \tag{24}$$

An unrecorded covariate whose aggregate correlation with the within-cell gain and the within-cell label distribution reaches 0.061 is enough to account for the whole of that $\widehat{\Delta R}$. At the $\bar{\rho} = 0.1$ that this section suggests as a plausible elicited value, the sharp bound is 0.02328, which *exceeds* the estimate. We therefore do not claim that a confounder would have to be implausibly strong to explain these Visual Genome gains away; on this evidence a modest one would suffice, and the reader should treat Appendix 5.4's covariance gains as conditional on the declared ϕ being close to complete. The flagship audit of Section 5.2 is not exposed in the same way, because its reported $\widehat{\Delta R}$ is indistinguishable from zero and a sensitivity bound on a null result changes nothing about it.

Table 3: Both forms of the Proposition 3 bound on the MLP-SPATIAL vs. MLP-CLASS pair, and the coarsening already reported in Table 7 ($K = 50$, $M = 2$ for the quadratic score). The empirical bias is the law-of-total-expectation aggregate of Eq. (19)'s covariance term across all subject-only cells. `experiments/sensitivity_bound_check.py` reproduces the free-bound rows from the committed ablation results and `experiments/sensitivity_analysis.py` the sharp rows from per-instance arrays.

Quantity	Value
$\widehat{\Delta R}$, ϕ = subject-only	0.02181
$\widehat{\Delta R}$, ϕ = class-pair	0.01439
Empirical coarsening bias	0.00742
Free bound, $M\sqrt{K-1}$, at $\bar{\rho} = 1$	14.00
Minimum $\bar{\rho}$ for the free bound to reach the bias	0.00053
Sharp bound B at $\bar{\rho} = 1$	0.23284
Sharp bound at $\bar{\rho} = 0.1$	0.02328
Robustness value $\rho^{\dagger}$	0.06127

4.7. *Exact finite-sample decomposition*

Consider cell c with $n_c \geq 2$ examples. Write $H_i(y) = H_{x_i}(y)$. WAGER computes

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_{i \in c} H_i(y_i), \tag{25}$$

$$\widehat{\Delta P}_c = \frac{1}{n_c(n_c - 1)} \sum_{i \neq j \in c} H_i(y_j), \tag{26}$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j \in c} A_{ij}, \tag{27}$$

where the antisymmetric kernel is

$$A_{ij} = \frac{1}{2}\{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \tag{28}$$

Theorem 3 (Finite-sample WAGER identity)**.** *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \tag{29}$$

Moreover, when the cell's examples are drawn i.i.d. from the cell population, $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for ΔR_c. Independence, not exchangeabil-

ity alone, is what unbiasedness requires: the independent-coupling definition of ΔP_c pairs an example with a label drawn independently from the same cell, so $\mathbb{E}[H_1(Y_2)]$ matches it only for independent within-cell draws.

The dataset-level quantities weight cells by their number of identified examples: $\widehat{\Delta Z} = N_*^{-1} \sum_{c:n_c \geq 2} n_c \widehat{\Delta Z}_c$ for $Z \in \{T, P, R\}$. Singleton cells cannot identify a within-cell counterfactual; WAGER excludes and reports them instead of inventing a smoothed estimate. If $\widehat{\Delta T} > 0$, the descriptive covariance share is $\widehat{\rho} = \widehat{\Delta R}/\widehat{\Delta T}$. It may exceed one when within-group covariance improves while the transported channel deteriorates, and it is undefined for a nonpositive total gain. We therefore report $\widehat{\Delta R}$ as the primary result, not a thresholded ratio.

4.8. Linear-time computation

Naively evaluating Eq. (26) is quadratic in cell size. Let m_{cy} be the count of label y in cell c. For example $i \in c$, its transported score is

$$P_i = \frac{\sum_{y=1}^{K} m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}. \tag{30}$$

Averaging P_i yields Eq. (26); averaging $H_i(y_i) - P_i$ yields Eq. (27). Thus the entire estimator costs $O(NK)$ time and $O(NK)$ storage for cached prediction vectors. No model fitting, projection fold, hierarchy, or smoothing hyperparameter appears.

4.9. Why leave-one-out transport, not a fitted-in-sample prior

Equation (30) excludes example i's own label from its transported score. It is natural to ask what is lost by the simpler alternative of fitting the cell's empirical label frequency on all n_c examples, including i, and using that fitted frequency as the counterfactual, i.e. treating $\hat{p}_c(y) = m_{cy}/n_c$ as a prior model and setting $\tilde{P}_i = \sum_y \hat{p}_c(y) H_i(y) = n_c^{-1} \sum_y m_{cy} H_i(y)$. This is the natural plug-in estimator, and it is what a frequency-collapsed projection baseline computes when no separate evaluation fold is withheld from fitting.

Proposition 4 (Exact attenuation of the in-sample plug-in)**.** *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq.* (30) *and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c}\hat{R}_i, \qquad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c}\hat{R}_i. \tag{31}$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_*^{-1}\sum_{c:n_c\geq 2}(n_c - 1)\widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out covariance gain by the factor $(n_c - 1)/n_c$.

The shrinkage in Eq. (31) is deterministic, not a variance statement: whenever $\widehat{\Delta R}_c \neq 0$, the in-sample plug-in is biased for it at every finite n_c, with the largest relative attenuation exactly in the smallest identified cells (50% at $n_c = 2$, vanishing only as $n_c \to \infty$). The standard remedy for this kind of self-prediction bias is a projection/evaluation split: a held-out fold restores unbiasedness by fitting $\hat{p}_c$ on data disjoint from the scored examples, but it does so by discarding examples from both stages and by introducing a split-fraction hyperparameter and, typically, higher variance. WAGER's leave-one-out U-statistic removes the bias exactly, at the full sample, for every ϕ-cell with $n_c \geq 2$, without a fitted nuisance distribution at all. Section 3.1 situates this result against the classical single-model score decomposition, where the analogous self-prediction bias affects each model's covariance estimate separately.

4.10. Inference

For point estimation we use all unordered within-cell pairs. For uncertainty, define $\bar{A}_i = (n_c - 1)^{-1}\sum_{j\neq i} A_{ij}$ and $\widehat{\Delta R}_c = n_c^{-1}\sum_i \bar{A}_i$. The estimated Hájek influence contribution for random cell membership is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \tag{32}$$

We sum $\widehat{\psi}_i$ within image and use the resulting one-way sandwich variance, so multiple relations from the same image are not treated as independent. Total-gain influence is $H_i(y_i) - \widehat{\Delta T}$; prior-gain and share intervals follow from the exact identity and delta method.

As a complementary finite-group diagnostic, WAGER applies an independently sampled cyclic shift to labels inside every cell. Each shift preserves cell size and label histogram. Recomputing $\widehat{\Delta R}$ under B shifts gives

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\widehat{\Delta R}^{(b)} \geq \widehat{\Delta R}\}}{B+1}. \tag{33}$$

This is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because SGG relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval; the randomization test is reported as a secondary relation-level diagnostic.

4.11. Asymptotic normality

Eq. (32) and the sandwich variance used for the reported intervals are introduced in Appendix B through a Hájek-projection expansion that is stated as $o_p(N_*^{-1/2})$ but not, until now, given a limit theorem. The following result supplies one, under regularity conditions that hold whenever the score is bounded and clusters do not grow without bound relative to the sample.

Theorem 4 (Asymptotic normality of the covariance-gain estimator)**.** *Suppose (i) $H_x(y) \in [-M, M]$ for all x, y; (ii) examples are partitioned into clusters $g = 1, \ldots, G$ (e.g. images) that are mutually independent, with cluster sizes having a finite third moment and $\max_g |g| = o(\sqrt{G})$; (iii) the limiting variance $\sigma^2 = \lim_{G\to\infty} \mathrm{Var}(\sqrt{N_*}(\widehat{\Delta R} - \theta))$ exists and is strictly positive, where θ is the identified-subpopulation estimand of Appendix B; and (iv) ϕ takes values in a fixed, finite set, each with strictly positive limiting population mass, so that every eligible cell's size n_c diverges almost surely as $N \to \infty$. Then*

$$\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{34}$$

and the plug-in sandwich variance $\widehat{\sigma}^2 = N_* \dfrac{G}{G-1} \sum_g \left(\sum_{i\in g} \widehat{\psi}_i / N_*\right)^2$ of Appendix B is consistent for σ^2.

Two things about condition (iv) deserve saying plainly rather than in a footnote. It is what licenses treating the per-cell plug-ins $\widehat{\Delta R}_c$ as consistent for ΔR_c in the variance argument, and a cell's size is not controlled by the aggregate identified count, so this needs its own divergence requirement. But it also forces every cell to be eventually identified, hence $N_*/N \to 1$: an earlier version of this theorem additionally assumed an identified fraction converging to some $\kappa \in (0, 1]$, which condition (iv) makes vacuous, and under which the identified-subpopulation estimand θ would in any case coincide with the unconditional target. We have therefore dropped that assumption rather than state two conditions that cannot both bind. The price is that the theorem does not describe the regime the paper's own applications sit in, where θ is a random, N-dependent target because coverage is short of one (99.0% on VG150, 98.9% in Section 5.2). What the theorem licenses is the interval's form and its variance estimator; what licenses reading the reported intervals at finite N is the coverage simulation of Section 5.1, and Section 6 states the gap between the two rather than eliding it.

The proof (Appendix A) combines the exact linearization of Appendix B with a Lindeberg–Feller central limit theorem for the independent-across-cluster sum of mean-zero, uniformly bounded terms [63, 64]; boundedness (i) makes the Lindeberg condition automatic given (ii), and consistency of $\widehat{\sigma}^2$ follows from the weak law of large numbers applied to each plug-in $\widehat{\Delta R}_c, \widehat{\Delta R}$ together with continuity of the variance functional. One step needs more care than the appendix's per-cell statement supplies: a per-cell Hájek remainder of order $o_p(n_c^{-1/2})$ does not aggregate to $o_p(N_*^{-1/2})$ at the dataset level when the number of cells grows, since summing C such remainders admits a $\sqrt{C}$ inflation. The repair does not need a new condition on cell sizes, only that the cell count be negligible relative to the identified sample, $\#\{c : n_c \geq 2\} = o(N_*)$: the second-order degenerate term of a two-sample U-statistic is $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$, and $\sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C) = o_p(N_*)$ gives the dataset-level rate directly. We state that condition explicitly; on VG150 it holds comfortably (C = 6,346 against N_* = 227,337). Condition (ii) is what rules out a single cluster dominating the sum – an image contributing a positive fraction of all relations, for instance – and is a standard requirement for cluster-robust inference generally, not specific to this estimator. This theorem licenses reading the image-clustered intervals reported throughout Section 5 as asymptotically valid in the idealized regime where every eligible cell's size grows without bound. That regime is not attained in-sample: VG150's eligible cells are heavy-tailed, and 2,286 of the 6,346 eligible class-pair cells hold fewer than five examples (Table 7). The proportion is smaller than earlier drafts of this paper asserted — those cells carry 2.7% of identified relations, and the median eligible cell holds appreciably more than the minimum — but a third of the cells still sit where condition (iv)'s divergence is plainly hypothetical. The reported 94.0% empirical coverage against a nominal 95% (Section 5.1), not this theorem directly, is therefore the evidence that the interval remains a good approximation well short of the idealization.

Relation to comparative forecast evaluation..

Corollary 3 (Relation to Diebold–Mariano/Giacomini–White)**.** *Under the trivial partition $\phi \equiv const$ (one cell containing every example), $\widehat{\Delta T} = N^{-1} \sum_i H_i(y_i)$ is the sample mean of the paired loss differential between the two models, and Theorem 4 applied to $\widehat{\Delta T}$ with its image-clustered interval is the grouped, cluster-robust form of the Diebold–Mariano statistic for that differential [25]; because the conditioning information set is empty, it is equivalently a degenerate instance of the Giacomini–White conditional predictive-ability test [26].*

This corollary is immediate from the definitions: DM and GW test whether a paired loss differential's (conditional) mean is zero, and $\widehat{\Delta T}$ at the trivial partition is exactly that differential's sample mean. $\widehat{\Delta P}$ and $\widehat{\Delta R}$ remain perfectly well defined at the trivial partition too — the single giant cell still supports leave-one-out label transport whenever $N \geq 2$ — but neither DM nor GW forms them: both tests target ΔT alone and stop there. At the trivial partition $\widehat{\Delta R}$ is the *unconditional* prediction–outcome covariance contrast — the Yates-type quantity of Section 3.1 with an empty conditioning set. It is not a classical resolution term: a resolution

Algorithm 1 WAGER gain decomposition

Require: frozen q_1, q_0; labels y; cells ϕ; proper score S

Ensure: $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty

$H_i(k) \leftarrow S(q_{1i}, k) - S(q_{0i}, k)$ for all i, k
for each cell c with $n_c \geq 2$ **do**
  count labels $m_{c1}, \ldots, m_{cK}$
  for each $i \in c$ **do**
    $P_i \leftarrow (\sum_k m_{ck} H_i(k) - H_i(y_i))/(n_c - 1)$
    $R_i \leftarrow H_i(y_i) - P_i$
  end for
end for
$\widehat{\Delta T} \leftarrow$ mean $H_i(y_i)$; $\widehat{\Delta P} \leftarrow$ mean P_i; $\widehat{\Delta R} \leftarrow$ mean R_i
compute image-clustered Hájek interval and cell-shift randomization p

term computed with no conditioning variable is identically zero, since it is the variance of a conditional mean given nothing, and $\widehat{\Delta P}$ is correspondingly not a calibration term. An earlier version of this paper asserted both identifications and they were wrong. What is true, and is the contribution at this special case, is the further exact split of the DM/GW differential into two components neither test forms. Conditioning on a nontrivial ϕ is what separates WAGER's estimand from the classical target more generally: DM/GW ask whether one forecaster beats another on average, optionally conditional on a test function of an information set [26]; WAGER asks the same question at every declared cell *and* attributes the answer, cell by cell, into a component recoverable from the label histogram alone and a component that is not. One qualification on how far this reading travels. Corollary 3 is stated at the trivial partition, and the statistic it identifies with a clustered DM test is $N^{-1} \sum_i H_i(y_i)$ over every example. The totals reported at a nontrivial ϕ in Section 5 are instead N_*-weighted means over the *identified* subsample, since cells of size one are excluded. Because ΔT itself does not depend on ϕ except through that exclusion, each reported $\widehat{\Delta T}$ is a clustered DM statistic for the paired differential *on its identified subsample* — which at the coverages reported here (99.0% on VG150, 98.9% in Section 5.2) differs from the full-sample DM statistic only in the excluded singletons. That is the precise claim; "exactly a clustered DM statistic" without the subsample qualifier, as earlier drafts had it in Appendix 5.4, is not right at a nontrivial ϕ.

5. Validation and an audit of a published method

Two questions have to be answered before the decomposition can be used on anything, and they are answered here in that order. Does the estimator recover what it claims to, with intervals that cover — and what, precisely, does it fail to separate? Section 5.1 answers both under controlled alternatives where the truth is known by construction. Then, does it say anything a benchmark's own metrics do not? Section 5.2 puts it to two publicly released checkpoints of a widely cited debiasing method, chosen so that the audited models are not ours and the claimed improvement is one the field already reports. Three further studies — our own Visual Genome predictors, a frozen-CLIP variant of them, and long-tailed image and text classification — apply the identical estimator and are reported in Section 5; All confidence intervals are 95%.

5.1. Controlled validation

We generate 24 cells over five labels, drawing each cell's label distribution from a Dirichlet law and taking the old model to be exactly the cell prior. The new model interpolates toward an oracle, $q_1 = (1-\beta)q_0 + \beta q_{\text{oracle}}$ with q_{oracle} placing 0.98 on the realized label, and forty repetitions are run at each $\beta \in \{0, .1, \ldots, .8\}$. Across that range the estimated covariance gain grows linearly and monotonically with the injected signal (Figure 3, left).

Type-I behaviour is assessed by perturbing the prior predictions with mean-zero noise and then randomly reassigning the perturbed vectors within each cell, which guarantees that no systematic prediction–label alignment survives. Over 200 such datasets the 99-shift randomization test rejects in 0.030 of runs at level 0.05, with a mean estimated covariance gain of -8.97×10^{-5}; the null p-values come out approximately uniform and the null estimates stay centred at zero (Figure 3, centre and right).

Coverage of the primary interval.. The oracle-signal and null designs above validate the estimator's centering and the randomization diagnostic, but the paper's primary uncertainty statement is the image-clustered Hájek interval, so we validate its coverage directly in the regime that stresses it: relations grouped into images (one to about thirty per image) whose oracle weight shares a latent per-image effect, and Zipf-distributed cell sizes in which most identified cells hold exactly two examples. Against the design expectation estimated from 4,000 independent replications, the cluster-robust interval covers in 94.0% of 500 runs, while an interval that wrongly treats relations as independent covers in only 88.6% – quantifying both the mild finite-sample undercoverage of the asymptotic interval and the necessity of clustering.

Discriminant validity.. Three further designs test that the channels separate what they claim to separate. First, a *calibration-only* alternative: the new model is a temperature-softened copy of an overconfident informative model, so the pair contains identical instance information. The raw decomposition credits it with a large negative covariance gain (mean −0.488) – the confound quantified on real models in Section 5.7 – while the calibration-matched protocol below reduces it to −0.0003 (sd 0.0017 over 200 runs), confirming that the protocol removes exactly the confidence-scaling channel. Second, a *prior-only* improvement (the new model corrects a heteroscedastically noisy estimate of the cell prior, adding no instance information) yields a covariance gain centred at zero (0.0002, sd 0.0035) with a clearly positive transported channel: prior improvements do not leak into $\widehat{\Delta R}$ beyond the algebraic guarantee of Theorem 1. Third, an *unrecorded within-cell shortcut* (a binary feature, varying inside each cell, that shifts the label distribution and is used by the new model) is credited to covariance (+0.041, sd 0.004), quantifying the stated limitation that $\widehat{\Delta R}$ measures all within-cell predictive signal relative to the declared ϕ, shortcut or otherwise, and cannot by itself distinguish the two.

The calibration-matched protocol.. The third of those designs identifies a limitation serious enough to be part of the procedure rather than a caveat about it. Because ΔR is a covariance between probability movements and labels (Theorem 1), rescaling a fixed model's confidence moves score between the two components while adding no information about any individual case: a monotone transformation of the predictions changes the decomposition even though it changes no ranking and no decision. We therefore adopt the following control wherever the compared models differ visibly in confidence. Split the evaluation units — images, where examples are clustered — at random into halves. On the first half, fit each model a single temperature T by maximum held-out likelihood on $q^{1/T}$ renormalized. On the second half, apply each model's own fitted temperature and run the decomposition there. The protocol costs nothing but a data split, it uses no label information from the audited half, and it is what the simulation above shows removes the confidence-scaling channel almost exactly.

How many splits each study averages over differs, and the difference is worth stating rather than glossing. The long-tailed classification study of Section 5.7 averages over twenty random splits. The audit of Section 5.2 and the matched rows of Section 5.6 each use a single split at a fixed seed. For those two the split is a source of variability the reported interval does not include, and it is not negligible: the calibration-only arm above has a between-split standard deviation of 0.001725, exceeding the half-width of the audit's own reported interval (0.00115). The audit's conclusion does not turn on this — its estimate is indistinguishable from zero and stays so under any split — but a reader comparing interval widths across our tables should know that two of the three matched studies report within-split uncertainty only. It does not make ΔR invariant to recalibration in general; it fixes the confidence scale at which the comparison is made, which is what a comparison of two *fixed* models needs.

5.2. Auditing a published debiasing method

The question the decomposition exists to answer concerns systems that other people release and that the field already compares, so the primary application is to a published pair rather than to predictors of our own. Tang et al. [7] release a causal MOTIFS PredCls checkpoint that yields two models from one set of

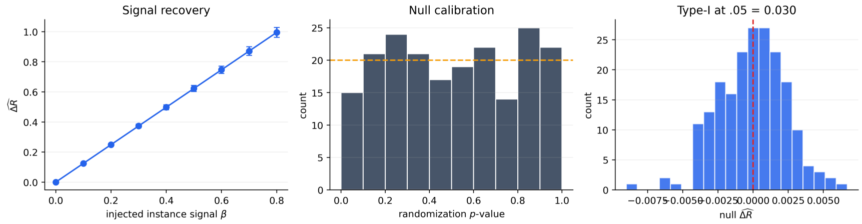

Figure 3: **Controlled validation.** WAGER recovers injected within-cell signal monotonically (mean ± one standard deviation), produces approximately uniform null randomization p-values, and controls type-I error in 200 null runs.

weights: the biased baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE), which subtracts a context-only counterfactual at inference. TDE is among the most cited debiasing methods in scene graph generation, and it is reported through mean recall, so it is exactly the kind of claimed improvement whose composition is at issue.

We evaluate both variants with the authors' own code on the canonical VG150 test split, which fixes the vocabulary, the split and the relation set, and makes the two prediction sets aligned relation-for-relation by construction. Our reproduction matches the published behaviour: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, and TDE trades the former for the latter, reaching $R@50 = 0.4588$ with $mR@50 = 0.2476$ and raising zero-shot recall from $zR@50 = 0.1102$ to 0.1432. Top-1 predicate accuracy falls from 0.6981 to 0.5577. The audit uses 183,639 relations over 26,446 images and 7,127 class-pair cells, of which 98.9% are identified.

What is audited, precisely.. One point of bookkeeping before the numbers, since it changes what the headline row is a statement about. The rows labelled calibration-matched do not decompose the released checkpoints as released; they decompose the two models after each has been rescaled by its own temperature ($T^* = 2.50$ for the baseline, 0.92 for TDE) fitted on a held-out half of the images. That is the comparison we argue is meaningful, because a proper score punishes miscalibration and these two models differ sharply in it. But the released pair, scored as it ships, gives a *significant* covariance loss (−0.01355, interval excluding zero), and a practitioner who downloads those weights and scores them is looking at the raw rows. We report both and label which is which.

Result.. Table 4 decomposes the pair. Read raw, TDE loses 0.11651 of quadratic score, of which 0.10296 sits in the transported channel and a small but significant 0.01355 in within-group covariance. That reading does not survive the control this paper insists on. The two models differ sharply in confidence — the baseline is overconfident at $T^* = 2.50$ while TDE, having subtracted a counterfactual, is already near-calibrated at $T^* = 0.92$ — and once temperatures are fitted on a held-out half of the images and the audit runs on the other half, the quadratic-score covariance channel is indistinguishable from zero: $\widehat{\Delta R} = -0.00006$ with a 95% interval of $[-0.00120, +0.00109]$ and a randomization p of .555. Essentially the entire proper-score difference, -0.18258 of -0.18264, is transported.

The log-score row of the same calibrated comparison does not repeat this null: it returns $\widehat{\Delta R} = +0.04396$, 95% CI $[+0.04079, +0.04714]$, an interval that excludes zero. The two proper scores therefore disagree on whether any real covariance change survives calibration matching – one reads exactly null, the other reads small but significantly positive – while agreeing that whichever value is correct is far smaller than the calibrated prior-channel shift (-0.18258 under the quadratic score, -0.59157 under the log score, both an order of magnitude or more larger than either covariance estimate). We do not adjudicate between the two scores here: nothing in Section 4.1 privileges one proper score's covariance estimate as the "true" one, and this is the reason the paper's default score is stated as a declared choice rather than derived. The claim this audit actually supports is the one both scores agree on — that TDE's proper-score improvement over the baseline is overwhelmingly transported *at the calibration-matched, class-pair configuration*, with a covariance remainder at most a small fraction of the total and possibly exactly zero — not the stronger, single-score claim that within-group covariance is unchanged.

That qualifier is load-bearing, and we would rather state its force than let a reader discover it in the table. Table 4's five configurations of this one pair give covariance gains of −0.21711, −0.12619, −0.01355, −0.00006 and +0.04396. At two of the five, “overwhelmingly transported” is simply false: the raw log-score row has $|\widehat{\Delta R}| = 0.126$ against $|\widehat{\Delta P}| = 0.159$, and the subject-only row has $\widehat{\Delta R} = -0.217$ dominating $\widehat{\Delta P} = +0.102$. We report all five and argue for one on stated grounds — matched confidence, because a proper score punishes miscalibration; the finest defensible cell, because Proposition 2 makes coarsening non-neutral — rather than presenting the favourable configuration as the finding. Theorem 2 is the reason the score-dependence is not a defect to be argued away: each Bregman generator defines its own exact estimand, so two proper scores disagreeing about ΔR are answering two well-posed questions, and which one an evaluator wants is a declaration, not a fact about the models.

This is a precise and, we think, fair characterization of what TDE does. Its own construction subtracts the context-only prediction, so a method that operates on the transported channel is what its authors built; WAGER quantifies that the intervention is almost purely of that kind, and that the model’s ability to tell which relation belongs to which image within a class-pair cell is left, at most, only marginally changed. The consequence for evaluation needs stating carefully, because it is easy to state in a way that is not quite true. Mean recall rose by 10.2 points; the quadratic-score covariance gain, calibration-matched at the class-pair cell, is −0.00006 with an interval containing zero. Those are different units and we have no bridge between them. Nothing here licenses the claim that ten points of mean recall bought 0.00006 of case-level evidence, and it would be equally wrong to say the mean-recall gain *is* transported, since the transported channel on the proper score *fell* over the same comparison ($\widehat{\Delta P} = -0.18258$). What the audit supports is a conjunction of two separate statements: the field’s headline metric moved ten points, and the proper-score difference between the same two models decomposes almost entirely into the transported channel, with a covariance remainder at most a small fraction of it. A reader may conclude that the metric movement is not evidence about within-cell case-level discrimination, because the quantity that would measure that barely moves. A reader may not conclude a

Table 4: WAGER audit of released checkpoints on the canonical VG150 PredCls split: MOTIFS-TDE against the MOTIFS-SUM baseline it debiases, both evaluated with the original authors' code. Raw rows use all identified relations; the calibration-matched rows fit each model's temperature on a held-out half of the images and audit the disjoint half. CI is image-cluster robust on $\widehat{\Delta R}$.

Regime	Score	Cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)
raw	quadratic	class pair	−0.11651	−0.10296	−0.01355 [−0.01504, −0.01206]
raw	log	class pair	+0.03277	+0.15897	−0.12619 [−0.13093, −0.12145]
raw	quadratic	subject	−0.11536	+0.10175	−0.21711
calibration-matched	quadratic	class pair	−0.18264	−0.18258	−0.00006 [−0.00120, +0.00109]
calibration-matched	log	class pair	−0.54760	−0.59157	+0.04396 [+0.04079, +0.04714]

conversion rate, and we do not supply one.

One further qualification: TDE is designed to change the decision rule rather than to emit calibrated probabilities, and a proper score evaluates the latter; the calibration-matched regime is what makes the comparison meaningful at all, and its necessity here is the clearest practical argument for the protocol of Section 5.1.

5.3. Relation prediction with controlled predictors

We work with Visual Genome [65] in the PredCls regime [66], where ground-truth boxes and object labels are supplied and the model has only to predict one of 50 predicate labels for each ordered object pair. The predictors of this section are audited on a VG150-*style* dataset we build directly from the released relationship annotations: the 150 most frequent object classes and 50 most frequent predicates form the vocabulary, and images are split 70/30 into a pool for fitting the compared predictors and a pool on which the audit runs. We use this reconstruction rather than the canonical VG150 release because these predictors are trained here from annotations alone; Section 5.2 audits *released third-party checkpoints* on the canonical split itself, so both settings appear in the paper and neither is asked to stand for the other. The reconstruction provides 532,987 training and 229,605 test relations, drawn from 65,228 and 27,955 images respectively, and the cell throughout is ϕ = (subject class, object class), two real examples of which appear in Figure 4. Of the 8,614 cells present in the test split, 6,346 contain at least two relations and are therefore identified; the remaining 2,268 are singletons, which WAGER excludes and reports rather than smoothing over, leaving 227,337 relations (99.0%) under audit.

Table 5: Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is transported.

Model	Accuracy
FREQ	0.6564
FREQ+OVERLAP	0.6545
MLP-CLASS	0.6554
MLP-SPATIAL	0.6639
MLP-SPATIAL+	0.6674

The five predictors are deliberately constructed so that each exposes a different kind of information. FREQ is the training-count distribution conditioned on the class pair, with add-one smoothing and a subject-marginal and then global backoff for class pairs unseen in training; FREQ+OVERLAP adds to it a single binary box-overlap indicator. MLP-CLASS consumes fixed subject and object class embeddings, which makes it a function of ϕ alone; MLP-SPATIAL extends those embeddings with 12 normalized box-geometry features covering relative centre, scale, aspect, intersection, containment and IoU, while MLP-SPATIAL+ both widens the network and trains longer (Appendix D lists every architecture and schedule), so its increment over MLP-SPATIAL reflects capacity and optimization jointly rather than capacity alone. Each is trained once on the training split, and WAGER thereafter consumes nothing but their cached test distributions, scored quadratically with image-clustered intervals and 499 within-cell cyclic randomizations, whose smallest attainable p-value is $1/500 = .002$; table entries of .002 are at this granularity floor and should be read as $p \leq .002$. With confidence intervals as the primary uncertainty statement and every significant effect many standard errors from zero, we do not additionally adjust the diagnostic p-values for the number of reported comparisons.

5.4. Gain attribution for the controlled predictors

The central result is collected in Table 6 and Figure 5. MLP-CLASS improves on FREQ by 0.03386 of quadratic score, yet because its gain vector is constant inside every class-pair cell, the estimator assigns that entire improvement to prior transport and precisely nothing to instance covariance. The zero is worth dwelling

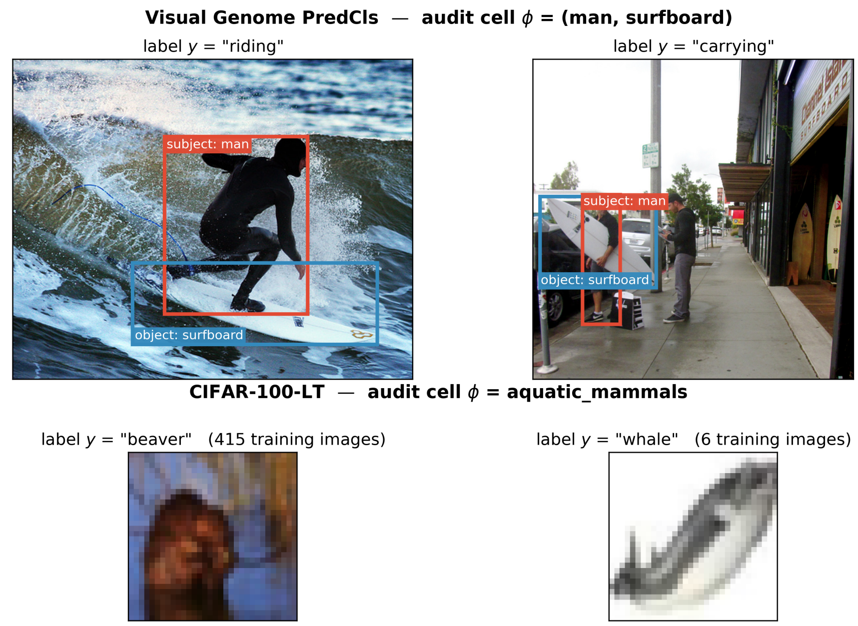

Figure 4: **What a cell contains, on both benchmarks.** Each row shows two real test examples that share the same declared grouping variable ϕ but carry different labels – precisely the configuration within-cell transport exploits. *Top:* two Visual Genome relations from the cell $\phi = (\text{man}, \text{surfboard})$. The class pair is identical and therefore cannot distinguish "riding" from "carrying"; only the image content can, which is what ΔR is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom:* two CIFAR-100 test images from the superclass ϕ = aquatic_mammals, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

on: it follows algebraically from the construction rather than from a learned baseline happening to pass a sanity check.

Adding box geometry changes the picture. MLP-SPATIAL gains 0.04270 over FREQ, of which 0.02832 survives label transport and 0.01439 is within-group covariance (CI $[0.01373, 0.01504]$). More revealing still is the direct comparison against MLP-CLASS, where the total gain falls to 0.00885 only because the transported channel deteriorates by 0.00554 even as the covariance gain from spatial features rises by 0.01439. The resulting covariance share of 1.63 records a genuinely useful instance-specific change that the aggregate score partly conceals behind a worse prior component, and it is exactly the configuration that a ratio confined to $[0, 1]$, or thresholded at 0.5, would misdescribe.

The remaining two comparisons bracket the interesting range. MLP-

Table 6: WAGER decomposition on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; p is the 499-shift relation-level randomization diagnostic (granularity floor .002). "–" denotes an undefined share because total gain is negative. The MLP-CLASS row's zero-width interval is exact by construction, not a sampling statement: a gain vector constant within every cell makes the antisymmetric kernel identically zero (Theorem 1), so no variability exists for the interval to measure.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	$\widehat{\rho}$	p
FREQ+OVERLAP	FREQ	−0.01229	−0.01430	0.00202 [0.00168, 0.00235]	–	.002
MLP-CLASS	FREQ	0.03386	0.03386	0.00000 [0.00000, 0.00000]	0.00	.724
MLP-SPATIAL	FREQ	0.04270	0.02832	0.01439 [0.01373, 0.01504]	0.34	.002
MLP-SPATIAL+	FREQ	0.04644	0.02885	0.01760 [0.01686, 0.01834]	0.38	.002
MLP-SPATIAL	MLP-CLASS	0.00885	−0.00554	0.01439 [0.01373, 0.01504]	1.63	.002
MLP-SPATIAL+	MLP-SPATIAL	0.00374	0.00053	0.00321 [0.00289, 0.00353]	0.86	.002

SPATIAL+ adds 0.00374 over MLP-SPATIAL, nearly all of it covariance gain at 0.00321 (CI [0.00289, 0.00353]), whereas FREQ+OVERLAP supplies the opposite caution: its overlap indicator does create a small positive covariance gain, but the accompanying negative prior term is larger, so the model ends up worse overall. Using image-derived information is thus never treated as an improvement in itself; both channels are reported, signs included.

Reading the magnitudes.. Quadratic-score units are not ones most readers calibrate by eye, so it is worth anchoring them in the rank-based metrics this literature reports. Where both channels point the same way the correspondence is straightforward: MLP-SPATIAL-S's $\widehat{\Delta R} = +0.01023$ over MLP-CLASS-S accompanies +0.51 points of top-1 predicate accuracy, +0.60 of mean reciprocal rank and +0.86 of recall@5, so in these data a hundredth of quadratic-score gain is worth roughly half an accuracy point. The covariance channel is not, however, a re-description of any rank metric, and the real-pixel comparison of Section 5.6 is where that becomes visible: there $\widehat{\Delta R} = +0.00641$ while top-1 accuracy falls by 3.07 points and recall@5 by 2.21. Prior-matching both models before scoring narrows the accuracy gap to 2.17 points without closing it. The rank metrics aggregate the two channels into one number; when the channels carry opposite signs, as they do here, no single such number can represent both, which is precisely the situation the decomposition exists to report.

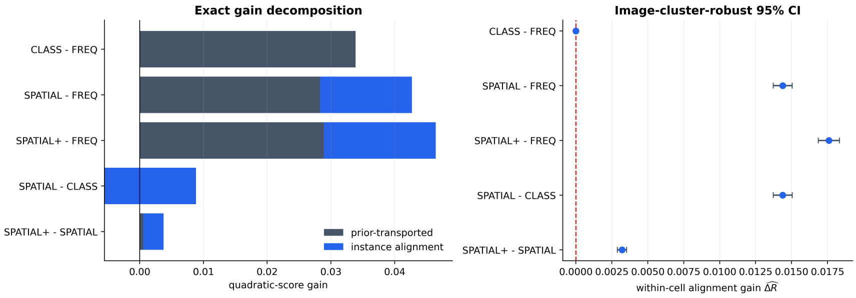

Figure 5: **Direct model-pair attribution.** Left: exact decomposition of each positive gain into transported and within-group covariance components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary covariance estimates with image-cluster-robust intervals.

Reading $\widehat{\Delta T}$ as a comparative forecast test.. Corollary 3 identifies $\widehat{\Delta T}$ and its image-clustered interval in Table 6 as a grouped Diebold–Mariano statistic and interval, on the identified subsample, for each pair's paired score differential, so every row already reports what that classical test would report. What it does not report is the next two columns: whether MOTIFS-TDE (Section 5.2) or MLP-SPATIAL beat their baseline for a reason a frequency-only remodeling of the same benchmark could not reproduce, which is the question $\widehat{\Delta P}$ and $\widehat{\Delta R}$ answer and DM/GW do not ask.

5.5. Sensitivity analyses

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table 7). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. The coverage column matters for reading the rows against each other, and earlier versions of this table omitted it. Each row targets the identified subpopulation its own setting defines, so the minimum-cell-size rows are not four estimates of one quantity: raising the minimum from 2 to 20 drops coverage from 99.0% to 86.7% and changes the estimand along with the estimator. That the estimate moves so little across them (0.01439 to 0.01225) is therefore mild evidence of stability rather than a sensitivity analysis holding the target fixed.

Coarsening ϕ from the class pair to subject class increases the credited covariance to 0.02181. Proposition 2 identifies why: the coarser cell's estimate equals

Table 7: Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on $\widehat{\Delta R}$.

Factor	Setting	$\widehat{\Delta T}$	$\widehat{\Delta R}$ (95% CI)	Cells	Cov.
Score	Brier (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Score	Log (floored)	0.04297	0.06536 [0.06294, 0.06779]	6346	99.0%
Cell ϕ	class pair (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Cell ϕ	subject only	0.00943	0.02181 [0.02079, 0.02283]	150	100.0%
Min. cell size	2 (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Min. cell size	5	0.00772	0.01395 [0.01329, 0.01462]	4060	96.3%
Min. cell size	10	0.00683	0.01330 [0.01262, 0.01398]	2744	92.6%
Min. cell size	20	0.00570	0.01225 [0.01155, 0.01295]	1757	86.7%

the class-pair estimate's conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to ΔP. This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot be assumed neutral, and only the finer partition's estimate isolates covariance gain net of every prior regularity expressible in ϕ. The same class-pair-versus-subject-only pair also instantiates Proposition 3 directly, with the subject class playing the role of the unrecorded confounder Z: Table 3 in Section 4.6 reports that the resulting bias of 0.00742 is comfortably inside the Cauchy–Schwarz bound, though the bound is loose enough at $K = 50$ that it would remain satisfied for a $\bar{\rho}$ three orders of magnitude smaller than one. Raising the minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. The procedure itself has no smoothing strength, split fraction, or decision threshold to tune.

5.6. Real pixels against a geometric proxy

None of the models in Sections 5.3–5.5 ever sees an image. They are functions of the declared class pair and, in the case of MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry serves only as a proxy for what an image contains, a sceptical reader is entitled to ask whether the covariance channel is responding to genuine pixel evidence or merely to that proxy.

Figure 4 sharpens the question: the cell ϕ = (man, surfboard) holds both "riding" and "carrying", and although their box configurations certainly differ, what actually distinguishes the two relations is what the photograph shows.

MLP-VISUAL is introduced to answer it. Subject and object regions are cropped from the original Visual Genome image and each is encoded by a frozen CLIP ViT-B/32 image encoder [67], after which the two 512-dimensional embeddings are concatenated with the class embeddings that MLP-CLASS already uses and passed to a small trained head of 256 and 128 units. Because CLIP itself is never fine-tuned, whatever covariance the model gains reflects information its pretraining had already recovered from pixels and the head has merely matched to an SGG label, rather than representation learning performed on Visual Genome.

Matched training data.. We report this comparison apart from Table 6 rather than as an additional row, for a reason that is itself an instance of the paper's argument. Encoding every training relation with CLIP is expensive enough that MLP-VISUAL must be trained on a subsample, and setting a subsampled model against the full-data MLP-SPATIAL would confound feature content with training-set size so thoroughly that a disappointing result could not honestly be attributed to the pixels. All three compared predictors are therefore retrained on a single fixed, seeded subsample of 100,000 of the 532,987 training relations, and are written MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S to mark the fact. Differing only in their feature sets, any gap between them isolates the features. The audit continues to use the complete 229,605-relation test split with the same class-pair ϕ and image-clustered intervals, so its numbers remain directly comparable with the main analysis. The sampling procedure, the crop-deduplication scheme and the count of images that could not be retrieved are given in Appendix D.

Result.. Judged by any aggregate measure, the CLIP model is the worst of the three. Its top-1 accuracy is 0.6161 against 0.6467 for MLP-SPATIAL-S and 0.6416 for MLP-CLASS-S, and its quadratic score is lower still, falling 0.04655 below MLP-SPATIAL-S. An evaluator working from either number would conclude that replacing box geometry with pixel content had simply made the predictor worse, and would discard it.

The decomposition in Table 8 and Figure 6 contradicts that reading on the axis the paper cares about. Against MLP-SPATIAL-S the CLIP model loses 0.05296 of transported gain while *gaining* 0.00641 of instance covariance (95% CI $[0.00514, 0.00769]$, randomization $p = .002$), so its entire deficit sits in the transported channel and its within-cell discrimination is significantly better, not worse. The comparison against the class-only baseline tells the same story more plainly: pixels buy 0.01665 of covariance where box geometry buys 0.01023, and the two intervals are far apart. Real image content therefore carries more instance-specific signal than the geometric proxy standing in for it, which is precisely what Section 5.6 set out to test.

Why the transported channel collapses has a plausible mechanism, though we state it as a hypothesis rather than a finding. The 1024 CLIP dimensions dominate the 64 dimensions of class embedding that the head also receives, so the model fits the class-pair frequency structure of VG150 markedly less well than a predictor built around that structure. A benchmark whose headline metric rewards transported gain will punish this model, and the aggregate numbers duly do. What WAGER adds is the observation that the punishment is not for failing to use the image; it is for failing to reproduce the annotation prior, while using the image better than either competitor.

The deficit and the advantage respond to different interventions.. If the two channels measure what they claim, they should respond asymmetrically to a post-hoc correction that supplies prior information but no instance information. We test this by prior matching: within each class-pair cell, a model's predictions are reweighted so that their cell mean matches the training-count histogram, a label-shift-style correction that uses no test labels. The correction registers, as it must, almost entirely in the transported channel ($\Delta P = +0.02401$ against $\Delta R = +0.00111$ for the CLIP model relative to its uncorrected self), and it cuts the CLIP model's aggregate deficit roughly in half (-0.04655 to -0.02143) while

Table 8: Matched-subsample real-pixel study on the full VG150 test split (coverage 99.0%, as in Table 6). All three predictors are trained on the same 100,000 relations and differ only in features. CI is image-cluster robust on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
MLP-SPATIAL-S	MLP-CLASS-S	0.00503	−0.00520	0.01023 [0.00967, 0.01079]	.002
MLP-VISUAL-S	MLP-CLASS-S	−0.04152	−0.05816	0.01665 [0.01534, 0.01796]	.002
MLP-VISUAL-S	MLP-SPATIAL-S	−0.04655	−0.05296	0.00641 [0.00514, 0.00769]	.002

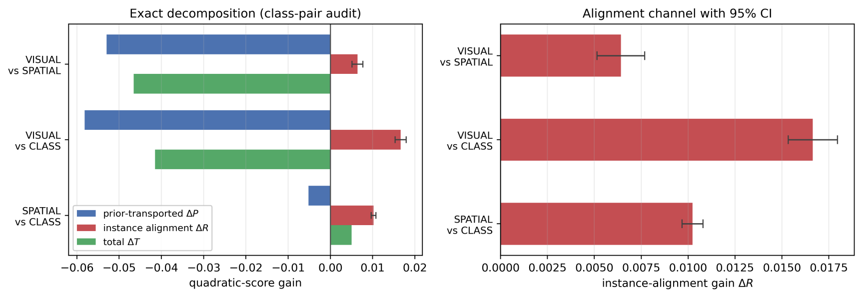

Figure 6: **Pixels beat geometry on covariance while losing on the aggregate.** Left: the exact decomposition for the three matched-subsample pairs. Both comparisons involving the CLIP model have strongly negative totals (green) driven entirely by the transported channel (blue), while their covariance channel (red) is positive in every case. Right: the covariance channel alone, with 95% image-clustered intervals. Replacing box geometry with real image content raises within-group covariance from 0.01023 to 0.01665 over the same class-only baseline.

leaving the covariance advantage intact (+0.00752, or +0.00663 with both models corrected). A calibration-matched audit – temperatures fitted on a held-out image split, $T^* = 1.18$ against 1.00, so these heads are nearly calibrated to begin with – likewise leaves the advantage significant at +0.00467 [+0.00314, +0.00621]. The covariance channel is therefore neither manufacturable by prior correction nor an artifact of confidence scaling. Conversely, the part of the deficit that prior matching removes is exactly the part a deployment could restore by re-weighting; the remainder reflects differences in within-cell prediction dispersion rather than in the fitted cell histogram.

This is the paper's argument running in the opposite direction from Section 5.7. There a near-zero total gain concealed a large covariance-gain *loss*; here a clearly negative total gain conceals a real covariance *gain*. In both cases the aggregate score is not merely imprecise but actively misleading about the thing a vision benchmark is meant to measure, and in both cases the two channels recover what it hides.

5.7. Long-tailed image classification

Sections 5.3–5.6 are all instances of one task (predicate classification on an ordered object pair). Section 3.4 argued that long-tailed image classification poses a structurally identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT [56], an exponential-imbalance (ratio 100) subsample of CIFAR-100's training split with the original, balanced test split retained for evaluation. We train ResNet-32 [68] classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) [56]; and the same re-weighting deferred to the final twenty epochs (DRW) [57]. Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Appendix D).

Choosing ϕ.. A declared grouping variable has to predict the label without determining it, and this second benchmark makes the constraint unusually easy to violate. Were ϕ taken to be the fine class label itself, every cell would be label-homogeneous, within-cell transport would have nothing to cross, and $\widehat{\Delta R}$ would come out identically zero – an algebraic artifact wearing the appearance of a finding. CIFAR-100's own 20 coarse superclasses avoid this, each gathering five fine labels into a cell of 500 balanced test images, which is structurally the situation of a VG class-pair cell holding several predicates (Figure 4). Alongside the superclass audit we report two further declared priors: the trivial coarsening to a single global cell, which audits against the global class marginal and so instantiates Proposition 2 in a second domain, and a partition into head, medium and tail training-frequency tiers. CIFAR-100 test images are independent, unlike the several relations VG draws from one photograph, so no cluster grouping is required and the plain Hájek interval of Section 4.10 applies directly. One semantic point matters in this domain: transport uses each cell's *test* label histogram, which on CIFAR-100-LT is balanced. A positive ΔP here therefore records movement *toward* the balanced test marginal – undoing the training-frequency skew – which is precisely the adjustment long-tailed evaluation is designed to reward; a transported gain is a description of where score was earned, not an accusation (Section 6).

Result.. The three classifiers reach top-1 accuracies of 0.3662 (CE), 0.2627 (CB) and 0.3874 (DRW), so re-weighting from initialization degrades the classifier substantially while the deferred schedule delivers the improvement it was designed to produce. Table 11 decomposes both gains against the superclass audit.

Of the two, the class-balanced run is the more instructive, precisely because its aggregate score conceals almost everything of interest. On the quadratic score it gains +0.00143 over the baseline, a difference small enough that a reader working from the headline number alone would reasonably conclude that re-weighting had changed very little. The decomposition tells another story: within superclasses the transported channel improves by +0.19713 while within-group covariance deteriorates by −0.19569 (95% CI [−0.20728, −0.18411]), so the near-zero total is not a small effect but two large ones that happen to cancel.

A calibration-matched regime.. A channel split of that shape could, however, be produced by confidence scaling alone. Under the quadratic score the covariance channel is a covariance between probability movements and labels (Eq. (12)), and a monotone recalibration moves it: softening an overconfident model transfers score mass from the covariance channel to the transported channel while adding no information about any individual image. The baseline invites exactly this concern, assigning a mean maximum probability of 0.764 while being correct on only 0.3662 of the test set. We therefore repeat the audit in a calibration-matched regime: each model's temperature is chosen by held-out likelihood on a calibration half of the test split ($T^* = 2.85$ for CE, 3.51 for CB, 2.61 for DRW), the decomposition is computed on the disjoint audit half, and the whole procedure is repeated over twenty random splits (Table 9). The regime's necessity is quantified by its control row: recalibrating the baseline alone – a transformation carrying no instance information – moves the channels by +0.380 (prior) and −0.175 (covariance), so the raw split does mix calibration with discrimination.

Table 9: Raw and calibration-matched decompositions on CIFAR-100-LT at the superclass audit. Temperatures are fitted by likelihood on a held-out calibration half of the test split; every decomposition uses the disjoint audit half; entries are means (sd) over twenty random splits. The final row is the control: recalibrating the baseline alone carries no instance information yet moves both channels substantially.

Comparison	Regime	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$
CB vs CE	raw	+0.00208 (0.00642)	+0.19635 (0.00369)	−0.19427 (0.00632)
CB vs CE	calibration-matched	−0.12912 (0.00373)	+0.11079 (0.00372)	−0.23991 (0.00491)
DRW vs CE	raw	+0.06712 (0.00865)	+0.04258 (0.00463)	+0.02454 (0.00685)
DRW vs CE	calibration-matched	+0.02173 (0.00275)	+0.00216 (0.00410)	+0.01957 (0.00329)
CE recalibrated vs CE	control	+0.20495 (0.00420)	+0.37960 (0.00268)	−0.17464 (0.00455)

The calibration-matched rows then separate the two schedules cleanly, and more sharply than the raw ones. CB's covariance deficit does not shrink but widens, to −0.23991 (0.00491): the model re-weighted from initialization has genuinely lost within-superclass discrimination – a real degradation that the ten-point accuracy drop records independently – and its near-zero raw total reflects that loss offset by its softer, better-calibrated confidence. DRW inverts the reading its raw split suggests. Raw, its +0.06712 gain divides roughly two-thirds transported to one-third covariance gain; calibration-matched, the improvement is +0.02173 (0.00275) of which +0.01957 (0.00329) is within-group covariance while the transported channel is close to zero (+0.00216 (0.00410)). What survives calibration matching is precisely the component the deferred schedule was designed to buy – per-instance discrimination, concentrated on rare classes (Table 12) – while the apparently transported share of its raw gain was largely the calibration difference between the two runs. Neither an accuracy difference nor an aggregate score exposes either composition, which is the practical argument for reporting the two channels, in both regimes, instead of their sum.

Seeds, imbalance ratios, and a post-hoc arm.. A decomposition computed from one trained pair carries only test-set sampling uncertainty, which is not the uncertainty relevant to a claim about a training *method*. We therefore retrained the triple at two further seeds at the primary ratio and at imbalance ratios 10 and 50, and added a fourth arm: post-hoc logit adjustment (LA), which divides the baseline's predictions by the training class prior without retraining [58]. Every configuration is audited in both regimes; Table 10 reports the calibration-matched covariance channel, which is the one the preceding paragraphs interpret.

Three things follow. First, the class-balanced run's covariance loss is not a misconfiguration but a systematic function of imbalance severity: -0.069 at ratio 10, -0.244 at 50, and -0.251 (0.011) across three seeds at 100, tracking an accuracy penalty that grows from 2.3 to 10.4 points over the same range. Second, the deferred schedule's covariance gain is positive at every ratio and in every seed, but its magnitude varies roughly fourfold across seeds at ratio 100 (+0.007 to +0.031). We therefore report the spread rather than a seed-level interval: the direction replicates, the magnitude is seed-sensitive, and the gap between the tight within-run intervals of Table 11 and this across-seed spread is exactly why a method-level claim needs replication rather than a single audit.

Third, LA is an instructive falsification of a natural expectation. A purely post-hoc prior correction might be expected to register as almost pure transported channel, as the within-cell prior matching of Section 5.6 does. It does not: calibration-matched, LA reads as almost entirely covariance gain (+0.022 to +0.038) with a transported channel near zero, while achieving the best balanced accuracy of any arm at every ratio. The reason is visible in the construction. Dividing by the *global* class prior is not a within-superclass histogram move, because fine classes inside a superclass differ in training frequency, and the correction it applies is multiplicative in each example's own predicted probabilities — it changes rankings where the model already carried latent evidence. The two post-hoc corrections in this paper therefore land in opposite channels, and correctly so: transport separates adjustments that act on the cell's label histogram from adjustments that act on individual predictions.

Where the covariance gain sits is visible in Table 12 and Figure 7 (raw regime, primary seed). For DRW it falls where the long-tail literature intends it to, reaching +0.09009 on few-shot classes and +0.06760 on medium-shot ones, while many-shot classes lose -0.07624 as the schedule shifts capacity away from the head. The class-balanced run reproduces that same ordering with every level displaced downward, its many-shot covariance collapsing to -0.41977. What the

Table 10: Calibration-matched covariance gain across seeds and imbalance ratios, with balanced test accuracy for each arm. LA is post-hoc logit adjustment applied to the baseline. The class-balanced deficit scales with imbalance severity; the deferred schedule's gain is positive throughout but seed-sensitive in magnitude; the post-hoc correction lands in the covariance channel rather than the transported channel.

		Top-1 accuracy				$\widehat{\Delta R}$ vs. CE (calibration-matched)		
Ratio	Seed	CE	CB	DRW	LA	CB	DRW	LA
10	0	0.5388	0.5154	0.5447	0.5497	−0.06903	+0.00617	+0.02222
50	0	0.4256	0.3242	0.4399	0.4550	−0.24409	+0.01903	+0.03768
100	0	0.3662	0.2627	0.3874	0.3959	−0.23936	+0.01987	+0.03449
100	1	0.3614	0.2287	0.3762	0.3877	−0.26176	+0.00748	+0.02672
100	2	0.3687	0.2481	0.3968	0.4024	−0.25213	+0.03124	+0.03601
100, mean (sd)						−0.25108 (0.01124)	+0.01953 (0.01188)	+0.03241 (0.00498)

Table 11: WAGER decomposition on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic, one-sided for a positive covariance gain, so the value of 1.000 on the CB rows records a covariance gain at the extreme negative end of its randomization distribution.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (20)	0.00143	0.19713	−0.19569 [−0.20728, −0.18411]	1.000
CB vs CE	global (1)	0.00143	0.25573	−0.25430 [−0.26801, −0.24058]	1.000
CB vs CE	tier (3)	0.00143	0.25247	−0.25103 [−0.26453, −0.23754]	1.000
DRW vs CE	superclass (20)	0.06606	0.04206	0.02400 [0.01340, 0.03460]	.002
DRW vs CE	global (1)	0.06606	0.03522	0.03085 [0.01860, 0.04309]	.002
DRW vs CE	tier (3)	0.06606	0.03702	0.02904 [0.01702, 0.04107]	.002

decomposition supplies, then, is not only how much of a long-tail gain is real but where in the frequency spectrum it was earned.

These numbers also provide an implementation check on real inputs. Computing the within-group covariance gain directly from the identity of Eq. (12), through a code path that shares nothing with the transport estimator, gives −0.19530; multiplying by the factor $n_c/(n_c - 1)$ that Proposition 4 predicts, at the $n_c = 500$ of a balanced superclass cell, returns −0.195693 against the estimator's −0.19569. The identities of Theorem 1 and Proposition 4 are algebraic, so this verifies the implementation rather than the theory; the point of the check is that two independent code paths agree to the fifth decimal on real trained models, not only on the synthetic populations used in the unit tests.

Coarsening behaves as Proposition 2 predicts. Moving from the 20 superclass cells to a single global cell changes the credited covariance from −0.19569 to −0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse audit overstates the covariance loss by attributing to it structure

Table 12: Covariance gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose covariance on the head.

		CB vs CE	DRW vs CE	
Tier	n	$\widehat{\Delta R}$	$\widehat{\Delta T}$	$\widehat{\Delta R}$
Few-shot (< 20)	3000	−0.03337	0.13785	0.09009
Medium-shot (20–100)	3500	−0.11075	0.10799	0.06760
Many-shot (> 100)	3500	−0.41977	−0.03740	−0.07624

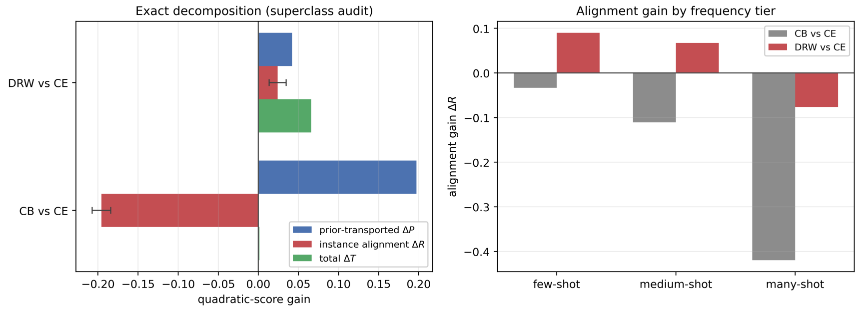

Figure 7: **The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at +0.00143 while its two channels reach ±0.2 – a near-zero aggregate score concealing two large opposing effects. DRW's smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the covariance channel by training-frequency tier, showing that DRW's genuine per-instance gain is concentrated on rare classes and negative on the head.

that the superclass partition correctly assigns to ΔP. The finer partition is again the conservative reading, for the same reason as in Section 5.5 and now in a second domain.

5.8. *Long-tailed text classification*

Sections 5.4–5.7 test two vision tasks that share cached probability vectors and grouping variables grounded in class frequency or object geometry. To test the breadth Section 6.3 claims – any benchmark supplying two models' full probability vectors, labels, and a declared discrete grouping variable – we apply the identical estimator, unmodified, to document category prediction on 20 Newsgroups [69]. Neither the predictors nor the grouping variable are visual here: documents replace images, and the declared ϕ is a topical grouping rather than a spatial or class-frequency one, so a favorable result cannot be attributed to anything specific to vision data.

We build a long-tailed training subset with the same exponential profile as CIFAR-100-LT (imbalance ratio 100, Section 5.7), applied to 20 Newsgroups' 20 fine categories, keeping the standard balanced test split (7,532 documents) untouched (Appendix D). The 20 categories fall into six conventional coarse topics – computing, recreation, science, politics, religion, and classifieds – which we use as the primary declared ϕ, the direct analogue of CIFAR's coarse superclasses; a global (trivial) and a training-frequency-tier partition are reported alongside, as in Section 5.7. Documents are represented by a TF-IDF vector reduced to 300 dimensions by truncated SVD, and the three compared classifiers – CE, CB, and DRW – follow the same three-arm protocol as Section 5.7: an identical small MLP architecture and optimizer across runs, differing only in per-example loss weight (Appendix D). Test documents are independent units, so the plain Hájek interval of Section 4.10 applies directly, as for CIFAR.

Result.. Table 13 decomposes both re-weighting schedules' gain over the cross-entropy baseline (top-1 accuracies 0.3532 CE, 0.3960 CB, 0.3776 DRW). The two schedules land in opposite channels here, which is itself the finding worth reporting: on this task it is class-balanced re-weighting from initialization, not the deferred schedule, that buys genuine within-topic covariance gain (+0.04317, CI $[0.03664, 0.04970]$, 37.9% of its total gain), while DRW's comparably sized total gain (0.11490 vs. CB's 0.11388) is 98.6% transported, its residual covariance statistically indistinguishable from zero at the superclass audit (+0.00155, CI $[-0.00296, 0.00605]$, $p = .236$) and significantly negative once the coarser tier partition is used instead (-0.00901, CI $[-0.01428, -0.00374]$). This is the reverse of Section 5.7's image-classification finding, where CB's near-zero total masked a large covariance-gain *loss* and DRW's gain was genuinely covariance-driven. Nothing about the estimator or the construction of ϕ changed between the two applications; what changed is which training recipe, on which data, buys real per-instance discrimination – exactly the composition question an aggregate score or an accuracy difference alone cannot answer, now demonstrated on a domain with neither a class-frequency table nor a spatial prior.

Table 13: WAGER decomposition on the 7,532-document balanced 20-Newsgroups-LT test split. Gains use the quadratic score; CI is the (unclustered) Hájek interval on $\widehat{\Delta R}$, documents being independent units; p is the 499-shift randomization diagnostic, one-sided for a positive covariance gain.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (6)	0.11388	0.07071	0.04317 [0.03664, 0.04970]	.002
CB vs CE	global (1)	0.11388	0.07203	0.04185 [0.03381, 0.04990]	.002
CB vs CE	tier (3)	0.11388	0.07810	0.03578 [0.02825, 0.04331]	.002
DRW vs CE	superclass (6)	0.11490	0.11335	0.00155 [−0.00296, 0.00605]	.236
DRW vs CE	global (1)	0.11490	0.11809	−0.00319 [−0.00884, 0.00245]	.980
DRW vs CE	tier (3)	0.11490	0.12391	−0.00901 [−0.01428, −0.00374]	1.000

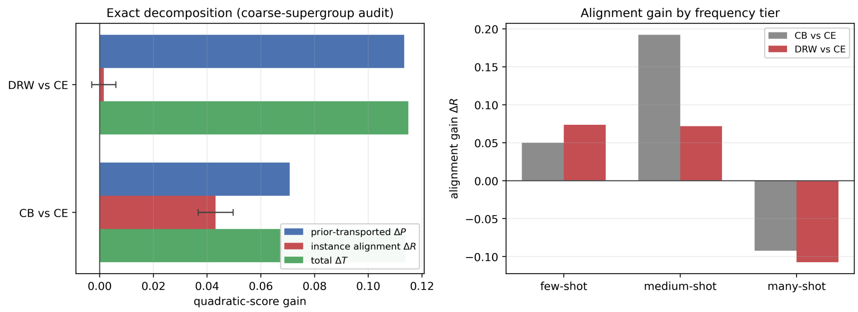

Figure 8: **A third domain, with neither a spatial nor a class-frequency prior.** Left: exact decomposition at the coarse-topic audit; CB's total gain is genuinely split between both channels while DRW's is almost entirely transported. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: covariance gain by training-frequency tier for both schedules, showing the same head-loses/tail-gains signature as Figure 7 despite the change of domain and grouping variable.

Coarsening moves both comparisons in the direction Proposition 2 allows without requiring: CB's credited covariance falls from 0.04317 at the superclass partition to 0.04185 at the global cell and 0.03578 at the tier partition, while DRW's changes sign twice, from a small positive residual (superclass) to a small negative one (global) to a significant negative one (tier). None of these differences overturns CB's conclusion, but DRW's illustrates exactly why the finest defensible partition is the conservative default (Section 4.5): a reader stopping at the coarser tier partition would report a significant covariance *loss* for a method whose finer-grained residual is not distinguishable from zero.

Table 14 breaks the superclass-audit covariance down by training-frequency tier for both schedules; the two methods agree in shape even though their superclass totals diverge sharply in composition, each gaining covariance on rare and medium-frequency categories and losing it on frequent ones, the same qualitative

Table 14: Covariance gain by training-frequency tier at the superclass audit, 20-Newsgroups-LT. Both schedules gain within-group covariance on rare and medium categories and lose it on frequent ones.

Tier	n	$\widehat{\Delta R}$, CB vs CE	$\widehat{\Delta R}$, DRW vs CE
Few-shot (< 20)	1954	0.04991	0.07352
Medium-shot (20–100)	2610	0.19213	0.07171
Many-shot (> 100)	2968	−0.09227	−0.10753

signature Section 5.7 reports for image classification.

This third domain uses a single trained pair per schedule rather than the multi-seed, calibration-matched protocol Section 5.7 applies to CIFAR-100-LT; extending that fuller protocol to text is the natural next replication, not a step this section claims to have taken.

6. Discussion and Limitations

6.1. What the decomposition does and does not establish

The unit of analysis is a model-pair gain rather than a model, and that change is what makes the two failure modes visible. A model may be a better estimator of the label frequencies within groups without predicting any individual case better, and be credited for the first as though it had achieved the second. Conversely, a model may resolve individual cases better while losing enough prior fit to hide the improvement in its aggregate score — Appendix 5.6 contains a case where the aggregate score and the accuracy both point the wrong way. The signed identity $\Delta T = \Delta P + \Delta R$ exposes both without a separately fitted reference forecaster, and the four-point example of Section 2 exhibits both in arithmetic.

The interpretation remains conditional on the declared grouping variable ϕ. A positive ΔR means the new model aligns its probability changes with the correct example better than the old model among examples sharing ϕ. It does not prove causal inference, compositional understanding, or robustness. Where this paper calls a covariance estimate "genuine," the word distinguishes a real statistical effect from a calibration or prior artifact; it is not a claim about reasoning, causal structure, or any compositional property of the model. If an unrecorded shortcut varies inside a cell, WAGER will count its predictive use in ΔR; Proposition 3 turns this qualitative caveat into a numeric one, bounding how far the reported covariance gain could be from what a fully adjusted audit would find as a function of a single, elicitable correlation parameter $\bar{\rho}$, though Table 3 shows that bound is conservative at VG150's label cardinality. Applications should therefore choose ϕ before looking at results, justify it from the benchmark's data generation process, and report finer and coarser alternatives. The observed increase under subject-only cells illustrates Proposition 2: coarsening adds a between-cell covariance term of either sign, so it cannot be assumed to leave the estimate unchanged, and only the finest identified partition isolates covariance gain net of every prior regularity expressible in ϕ. This is why the finest defensible prior feature is the conservative primary choice, not merely an empirical convention.

Two further interpretation limits deserve explicit statement. First, ΔR is not invariant to monotone recalibration: because the covariance channel is a covariance between probability movements and labels, sharpening or softening a fixed model shifts score mass between the two channels while adding no information about any individual example. Appendix 5.7 quantifies the effect on real models and adopts the corresponding control – decompose calibration-matched model pairs, with temperatures chosen on data disjoint from the audit – which we recommend whenever the compared systems differ visibly in confidence. Second, a transported gain is not evidence of anything illegitimate. Fitting the deployment label distribution better is genuine, decision-relevant skill whenever that distribution matches the benchmark's; it is fragile precisely under label shift, where transported gain can be added or removed post hoc by re-weighting [14, 15], as the prior-matching experiment of Appendix 5.6 demonstrates directly. The decomposition reports where a gain lives; whether the transported channel is worth paying for is a property of the deployment, not of the model pair.

Fineness is not a total order, and the guidance needs that qualification.. The advice to prefer the finest defensible ϕ is sound against *coarsening*, which is what Proposition 2 governs. It gives no guidance at all between two partitions of the same granularity, neither of which coarsens the other — and there the estimand can differ without limit. Four cases suffice. Take two labels, q_0 uninformative at $(\frac{1}{2}, \frac{1}{2})$, and a new model that is confidently right on all four: $q_1 = (0.9, 0.1)$ on the two cases labelled 1 and $(0.1, 0.9)$ on the two labelled 2. The total gain is $\widehat{\Delta T} = +0.480$ under every partition, and every partition below is fully identified at 100% coverage:

declared ϕ	cells	$\widehat{\Delta P}$	$\widehat{\Delta R}$
$\{1,3\},\{2,4\}$ (labels differ within each cell)	2×2	-1.120	$+1.600$
$\{1,2,3,4\}$ (one cell)	1×4	-0.587	$+1.067$
$\{1,2\},\{3,4\}$ (labels agree within each cell)	2×2	$+0.480$	0.000

The first and third partitions are equally fine, both fully identified, and they disagree about whether this model pair's improvement is entirely case-level or entirely transported. The third is the degenerate case the paper already warns about — grouping by something that determines the label leaves each cell label-homogeneous and drives $\widehat{\Delta R}$ to zero algebraically — but the warning was previously attached to the choice $\phi = Y$, and the point here is more general and less comfortable: any ϕ correlated with the label inside a cell partially degenerates the estimand, "finest defensible" cannot detect it, and Proposition 2 is silent because neither partition is a coarsening of the other. What the analyst actually needs to declare, and what we recommend reporting alongside ϕ, is the within-cell label heterogeneity that remains: a partition whose cells are nearly label-homogeneous cannot identify much of anything, and the coverage statistic does not reveal it. This is a limitation of the construction rather than of any application of it, and we do not have a procedure that resolves it.

A recalibration-invariant alternative, and what it would cost.. Because ΔR is a covariance between forecast values, its sensitivity to recalibration is structural rather than incidental, and a reader may reasonably ask why we do not use a rank-based within-cell contrast instead — a within-cell paired AUC difference, say, for which DeLong et al. [24] already supply the correlated-U-statistic variance.

Such a quantity would be invariant to any monotone rescaling of either model and would dissolve the confound this paper spends a protocol on. We think the trade is worth stating and is not obviously in our favour on every axis. What a rank-based contrast gives up is the exact identity: it is not a component of the score difference, so it cannot be added to anything to recover ΔT, and a benchmark reporting it learns how the models rank cases but not how the reported score gain decomposes. The construction here buys exact additivity with respect to a declared proper score, and pays for it with scale dependence. A rank-based diagnostic is the natural companion rather than a competitor, and reporting both — one exact and scale-dependent, one invariant and non-additive — is what we would now recommend to a benchmark that could choose.

"Choose ϕ before looking" is stated advice, not an enforced mechanism, and the analyst-chosen-subgroup problem has structural answers that WAGER should inherit. Our recommendation is that the *benchmark*, not the submitting author, declare the grouping variable – the annotation prior is a property of the dataset, and maintainer-declared ϕ removes the incentive to report only the most favorable of several candidates. We should be candid that this only moves the discretion rather than eliminating it. Maintainers are not disinterested: they are often the authors of methods evaluated on their own benchmark, sometimes a single unfunded volunteer, and sometimes absent, and a maintainer-declared ϕ that happens to flatter the benchmark's own baseline is no better than an author-declared one. The recommendation also sits in slight tension with our own advice to report every pre-declared candidate: if all candidates are reported, the declaration matters less, and if one is privileged, the choice matters more. What we would defend is the weaker claim that ϕ should be fixed and public *before* a leaderboard opens, by whoever does it, so that the partition is not a degree of freedom exercised after seeing results. Where several defensible priors exist, all pre-declared candidates should be reported, as our sensitivity tables do; guarantees that hold uniformly over a class of groupings are the subject of multicalibration [27], and porting that style of guarantee to pairwise gain attribution is a natural extension. To make reports comparable across papers, we suggest a fixed reporting block per audited pair: the declared ϕ and its rationale, the identified coverage, both channels signed with the covariance interval, the calibration-matched row when the models differ visibly in confidence, and the granularity sensitivity rows.

Exact transport also has a positivity requirement: a cell needs at least two examples. WAGER reports the identified fraction and makes no claim for singletons. Very fine ϕ can therefore improve confound control while reducing coverage. One further subtlety: cells identified at a fine granularity need not coincide with those at a coarse one, so cross-granularity comparisons mix the between-cell covariance term of Proposition 2 with a change of identified subpopulation – immaterial at VG150's 99.0% coverage, but worth restricting to a common identified subsample when coverage is low. Approximate matching could extend the method to continuous priors, but would replace the exact identity's clear intervention with modeling assumptions; we leave that extension open.

The image-clustered confidence interval is asymptotic, and Theorem 4 is what justifies that: under a bounded score, a mild condition ruling out one cluster from dominating the sample, a cell count negligible against the identified sample (Eq. (B.1)), and a fixed, finite ϕ-support whose every cell's size diverges in the limit, the dataset-level estimator is asymptotically normal around the identified-subpopulation estimand, with a consistent sandwich variance. The last condition is an idealization no heavy-tailed application attains in-sample. Earlier drafts of this paper put the point too strongly, asserting that most eligible VG150 cells sit at the minimum size $n_c = 2$; that is not so, and the correction runs in the paper's favour. Of 6,346 eligible class-pair cells, 2,286 hold fewer than five examples and those carry 2.7% of identified relations, so the bulk of the evidence comes from cells appreciably larger than the minimum. A third of the cells nonetheless sit where the divergence condition is hypothetical, which is why the reported 94.0% empirical coverage against a nominal 95% (Section 5.1) remains the operative evidence for the interval's finite-sample adequacy, with the theorem supplying the limiting justification the simulation checks against. The cyclic randomization test is finite-group exact only under its sharper within-cell exchangeability null and treats relations as the randomized units, so it is secondary in the SGG analysis. A future image-level randomization design that preserves several relation-cell margins simultaneously would strengthen finite-sample inference for structured benchmarks.

The audit of Section 5.2 indicates what adoption actually requires. Two released checkpoints, evaluated with their authors' own code, were enough: the method consumed the resulting probability arrays and returned a decomposition with intervals in seconds. Nothing about that workflow is specific to scene graph generation, and the practical obstacle is neither compute nor complexity but disclosure — benchmarks overwhelmingly collect ranked predictions rather than predictive distributions. A benchmark that archived per-example probabilities alongside its leaderboard would make this kind of audit routine for every submitted pair. Where several grouping variables are defensible, we would further recommend that the benchmark, rather than the submitting author, declare ϕ, and that all pre-declared candidates be reported.

Finally, WAGER evaluates probability vectors, not only top-1 decisions. This is a feature—the method can detect useful redistribution below the argmax—but it requires released logits or probabilities. The quadratic score is bounded and strictly proper; other proper scores answer scale-dependent variants of the same question and should be declared in advance.

6.2. Where else this construction lives

Decomposing a difference between two groups or systems into a part attributable to composition and a part attributable to conditional behaviour is not specific to forecast evaluation, and a reader arriving from another field will recognize the shape of $\Delta T = \Delta P + \Delta R$ immediately. In labour economics, the Oaxaca–Blinder decomposition splits a mean wage gap into an endowments component and a coefficients component [70, 71], and its semiparametric successors reweight one group's covariate distribution to another's [72] — which is what within-cell label transport does, in the other direction, to labels rather than covariates. In epidemiology, the Mantel–Haenszel estimator aggregates a within-stratum association across strata precisely so that composition cannot drive it [73], and the observation that a within-stratum association need not agree with the marginal one is non-collapsibility — of which Proposition 2's between-cell covariance term is an instance. Psychometrics asks a structurally identical question of test items: differential item functioning tests whether an item behaves differently across groups *conditional on ability*, and does so through a Mantel–Haenszel statistic [74]. Clinical prediction has long separated discrimination from calibration for a single model, which is the single-forecaster ancestor of the split this paper takes for a pair.

We claim no technical novelty from these connections and have not adapted any of their machinery. They are worth stating for two reasons. They locate the construction for readers who would otherwise see only a scoring-rule paper, and they suggest where the open problems already have answers: the analyst-chosen-partition problem this section worries about is the multiple-comparisons problem those literatures have faced for decades, and the label-shift exposure of Section 2.1 is a reweighting problem DiNardo et al. [72] solve in a setting where the weights are estimated rather than known.

6.3. Broader applicability

Four aligned arrays are all the algorithm requires – two models' probability vectors, the labels, and the prior-cell identifiers – and nothing in Sections 4.1–4.10 is specific to relation prediction. Appendix 5.7 puts that generality to the test on long-tailed image classification, where taking the benchmark's coarse superclasses as ϕ lets the decomposition distinguish a re-weighting schedule whose near-zero aggregate score hides two large cancelling channels from one whose genuine improvement is part prior-refitting and part rare-class covariance. The claim of breadth thereby acquires empirical content instead of resting on analogy. That second domain is also where the constraint on ϕ shows its teeth, since the fine class label cannot serve as a cell: it would leave every cell label-homogeneous and drive ΔR to zero by construction, producing an artifact that reads exactly like a null result. Elsewhere the natural choices are easy enough to name – a predeclared question template for VQA, an answer format or prompt family for other multimodal benchmarks – though we have not evaluated them, and we flag them as the obvious next domains rather than claim a coverage we have not earned. Whatever the setting, the scientific claim has to stay precise, because what WAGER measures is new prediction–label alignment within the declared cells and nothing more. Its value lies not in any single ϕ settling what counts as understanding, but in making every attribution an explicit and reproducible counterfactual tied to a declared nuisance structure, one that the same unmodified construction has now answered sensibly on two structurally different tasks.

7. Conclusion

When two probabilistic classifiers are compared on a benchmark whose labels depend partly on a feature both models observe, the reported score difference answers a question nobody asked. It sums a closer fit to the label frequencies within groups and a closer match between predictions and the individual cases they were made for, and those are different achievements with different consequences: the first is real skill that a lookup table also supplies and that a shift in label frequencies removes, the second is not.

This paper separates them exactly. Scoring each model's prediction for one test case against the label of a different case in the same group leaves the group and its label frequencies intact while breaking the correspondence between prediction and case, and the gain that survives that relabelling is the transported component. The identity $\Delta T = \Delta P + \Delta R$ then holds in the observed sample, for any score, with no fitted reference model, no held-out fold and no tuning parameter, at a cost of $O(NK)$. The remainder is a U-statistic of order two whose value is a within-group covariance between the two models' probability change and the label indicator — for every Bregman score, so the quadratic and logarithmic cases are one result rather than two (Theorem 2). That covariance is the object a covariance rearrangement of a proper score isolates [10], evaluated for a pair of forecasters instead of one. It is not the resolution term of the classical reliability–resolution partition, which is invariant to monotone rescaling of the forecast where a covariance between forecast values is not; the two coincide only under within-cell calibration, and Section 3.1 states the discrepancy. Naming the quantity correctly is what makes the recalibration sensitivity predictable rather than surprising, and it is why confidence matching is part of the procedure here.

What makes it usable rather than merely exact is that the pairwise contrast comes with its own inference. The influence function of the difference gives cluster-robust intervals in one pass; the estimator is asymptotically normal with a consistent sandwich variance (Theorem 4); and at a trivial grouping variable the undecomposed statistic is a clustered Diebold–Mariano test (Corollary 3), so every comparison reported here already contains what that classical test would say, plus the split it does not provide. Four exact statements settle design choices that would otherwise be judgment calls: using a group's own label frequencies as the counterfactual attenuates the estimate by precisely $(n_c - 1)/n_c$, which is why no sample splitting is needed (Proposition 4); merging groups moves the estimand by an explicit between-group covariance of either sign, so a coarser grouping cannot be assumed harmless (Proposition 2); the transported component splits exactly into a mean-forecast-fit term minus a within-cell dispersion term, so it is not a measure of prior fit and we do not call it one (Proposition 1); and a Cauchy–Schwarz bound limits how far an unrecorded grouping variable could move the result, from one elicited correlation (Proposition 3). On that last point we report the bound against our own numbers: for the Visual Genome pair of Appendix 5.3 an unrecorded covariate correlated at 0.061 would account for the entire reported gain, so those covariance gains are conditional on the declared grouping being close to complete.

The controlled experiments are as informative about the limits as about the capability. Prior-only improvements stay out of the covariance component, and injected within-group signal is recovered monotonically with intervals that cover at 94.0% against a nominal 95% in the clustered, small-cell regime that stresses them hardest. But unrecorded within-group signal is credited to resolution, shortcut or not, and a monotone recalibration moves score between the two components while adding no information — which is why calibration matching is part of the procedure and not a caveat attached to it.

Applied to two released scene-graph checkpoints, evaluated with their authors' own code, the decomposition gives a precise account of a widely cited debiasing method: its ten-point mean-recall improvement is, once calibration is matched, overwhelmingly transported under either of two proper scores, with a covariance remainder the two scores disagree about — indistinguishable from zero under one, small but significant under the other — and in both readings at least an order of magnitude smaller than the prior-side shift. Since the method's own construction subtracts a context-only prediction, this characterizes what it does rather than indicting it. The consequence for evaluation is the paper's thesis in a published setting: a metric the field reads as evidence about rare predicates can move ten points while case-level evidence changes by at most a small fraction of that, possibly none. Three further studies, on predictors whose information content is controlled by construction and in two other data modalities (Section 5), show the same estimator returning sensible attributions without modification, including a case in which a model that loses on every aggregate measure is shown to resolve individual cases significantly better than the model it appears to lose to.

Two limits deserve to be carried out of the paper alongside the construction. The estimand is not determined by the granularity of the grouping variable: two equally fine, fully identified partitions of the same four cases can split the same total gain as entirely case-level or entirely transported (Section 6), and the advice to choose the finest defensible ϕ cannot arbitrate between them because neither coarsens the other. And a rank-based within-cell contrast would be invariant to the recalibration this construction is sensitive to, at the cost of the exact additivity that is the point here; the two are companions rather than competitors, and a benchmark able to report both should.

The estimator's requirements are its main obstacle to routine use, and they are about disclosure rather than difficulty: it needs both models' full probability vectors, and benchmarks overwhelmingly archive ranked predictions instead. A leaderboard that stored per-example probabilities, and whose maintainers declared the grouping variable rather than leaving it to submitting authors, would make this kind of attribution available for every pair it ranks.

Data and code availability

All code, cached model outputs and scripts needed to reproduce every number, table and figure in this paper are publicly available at `https://github.com/vinhqdang/WAGER`. The repository contains the estimator, its unit tests, every experiment driver, and the cached results files that each reported value is checked against by an included verification script. The two audited scene-graph checkpoints are the public release of Tang et al. [7]; Visual Genome [65] and CIFAR-100 are public benchmarks.

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A. Proofs

A.1. Proof of Theorem 1

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y)\mathbb{1}\{Y = y\}], \qquad \Delta P_c = \sum_y \mathbb{E}[H_X(y)] \Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (11); the first identity is the definition of ΔR_c. Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot \mid X)\|_2^2 - \|q_0(\cdot \mid X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over y is $\mathrm{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \mathrm{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (12). If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. □

A.2. Proof of Proposition 1

Condition on $C = c$ and let $Y' \sim p(c)$ be independent of X. For the quadratic score $S_\mathrm{B}(q, y) = 2q_y - \|q\|^2$,

$$\mathbb{E}[S_\mathrm{B}(q_m, Y') \mid c] = 2\langle p(c), \bar{q}_m(c)\rangle - \mathbb{E}[\|q_m\|^2 \mid c]. \tag{A.1}$$

Decompose the second moment as $\mathbb{E}[\|q_m\|^2 \mid c] = \|\bar{q}_m(c)\|^2 + \operatorname{tr}\operatorname{Var}(q_m \mid c)$, and complete the square using $2\langle p, \bar{q}\rangle - \|\bar{q}\|^2 = \|p\|^2 - \|p - \bar{q}\|^2$:

$$\mathbb{E}[S_{\mathrm{B}}(q_m, Y') \mid c] = \|p(c)\|^2 - \|p(c) - \bar{q}_m(c)\|^2 - \operatorname{tr}\operatorname{Var}(q_m \mid c). \tag{A.2}$$

Subtracting the $m = 0$ expression from the $m = 1$ expression, the $\|p(c)\|^2$ terms cancel and Eq. (13) follows, since $\Delta P_c = \mathbb{E}[S_{\mathrm{B}}(q_1, Y') \mid c] - \mathbb{E}[S_{\mathrm{B}}(q_0, Y') \mid c]$ by definition. □

A.3. Proof of Theorem 2

Condition on $C = c$. The proof of Theorem 1 above establishes $\Delta R_c = \sum_y \operatorname{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)$ for *any* score contrast H, not only the quadratic one; only the identification of $H_x(y)$ with a specific functional form is score-dependent. For the Bregman score of Eq. (14),

$$H_x(y) = \underbrace{[\psi(q_1(x)) - \psi(q_0(x))] - [\langle\nabla\psi(q_1(x)), q_1(x)\rangle - \langle\nabla\psi(q_0(x)), q_0(x)\rangle]}_{=:B(x)} + \Delta g_y(x),$$

writing $q_i(x)$ for $q_i(\cdot \mid x)$, where $B(x)$ collects every term not indexed by y and $\Delta g_y(x) = \partial_y\psi(q_1(x)) - \partial_y\psi(q_0(x))$ is the y-indexed remainder, exactly as in the proof of Theorem 1 for the quadratic case. Substituting,

$$\begin{aligned}\sum_y \operatorname{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid c) = \operatorname{Cov}\Big(B(X), \sum_y \mathbb{1}\{Y = y\} \Big| c\Big)& \\ + \sum_y \operatorname{Cov}(\Delta g_y(X), \mathbb{1}\{Y = y\} \mid c).&\end{aligned}$$

The first term is $\operatorname{Cov}(B(X), 1 \mid c) = 0$ since $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The second term is Eq. (16). □

A.4. Proof of Theorem 3

Summing the kernel in Eq. (28) over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c - 1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c}\sum_i H_i(y_i) - \frac{1}{n_c(n_c-1)}\sum_{i\neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. When the cell's examples are i.i.d. draws from the cell population, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation [35]; independence is used, not merely exchangeability, because ΔP_c is defined through an independent label coupling, so $\mathbb{E}[H_1(Y_2)] = \mathbb{E}[H_X(Y')]$ requires the two draws to be independent. □

A.5. Proof of Proposition 2

Fix y and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable C nested inside $\bar{c}$ gives

$$\begin{aligned}\mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid \bar{\phi} = \bar{c}) &= \mathbb{E}[\mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C) \bigm| \bar{\phi} = \bar{c}] \\ &\quad + \mathrm{Cov}(\mathbb{E}[H_X(y) \mid C], \Pr(Y = y \mid C) \bigm| \bar{\phi} = \bar{c}),\end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} \mid C] = \Pr(Y = y \mid C)$. Summing over $y = 1, \ldots, K$ and applying the population identity of Theorem 1 to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over c within $\bar{c}$) yields Eq. (18). □

A.6. Proof of Proposition 4

Fix a cell c with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^{K} m_{cy} H_i(y) = \sum_{j\in c} H_i(y_j) = H_i(y_i) + \sum_{j\neq i} H_i(y_j)$, and Eq. (30) gives $\sum_{j\neq i} H_i(y_j) = (n_c - 1)\hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c}\sum_y m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1)\hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1}\hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1}\hat{R}_i = (n_c - 1)n_c^{-1}\hat{R}_i$. Averaging $\tilde{R}_i$ within cell c and reweighting by n_c across cells, as in the dataset-level definition of $\widehat{\Delta Z}$, gives $\widetilde{\Delta R} = N_*^{-1} \sum_c n_c \cdot \frac{n_c-1}{n_c} \widehat{\Delta R}_c = N_*^{-1} \sum_c (n_c - 1)\widehat{\Delta R}_c$. $\square$

A.7. Proof of Proposition 3

Fix $\phi = c$ throughout and write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$, both random through their dependence on (ϕ, Z). Let $U = (a_y - \mathbb{E}[a_y \mid c])_{y=1}^K$ and $V = (b_y - \mathbb{E}[b_y \mid c])_{y=1}^K$ be the corresponding mean-zero $\mathbb{R}^K$-valued random vectors conditional on $\phi = c$. The map $(U, V) \mapsto \mathbb{E}[\langle U, V\rangle \mid c] = \sum_y \mathrm{Cov}(a_y, b_y \mid c)$ is an inner product on the space of mean-square-integrable $\mathbb{R}^K$-valued random variables conditional on $\phi = c$, so by the Cauchy–Schwarz inequality applied to that inner product,

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \leq \sqrt{\mathbb{E}[\|U\|^2 \mid c]}\, \sqrt{\mathbb{E}[\|V\|^2 \mid c]} = \sqrt{\textstyle\sum_y \mathrm{Var}(a_y \mid c)}\, \sqrt{\textstyle\sum_y \mathrm{Var}(b_y \mid c)}.$$

By the definition of ρ, the left side equals $|\rho| \sqrt{\sum_y \mathrm{Var}(a_y \mid c)} \sqrt{\sum_y \mathrm{Var}(b_y \mid c)}$ exactly, so this step alone is not yet a bound; two further steps supply one, at two different levels of conservatism. First, $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$ is itself a conditional expectation of $H_X(y)$ given more information than $\phi = c$ alone, so the law of total variance gives $\mathrm{Var}(H_X(y) \mid c) = \mathbb{E}[\mathrm{Var}(H_X(y) \mid \phi, Z) \mid c] + \mathrm{Var}(a_y \mid c) \geq \mathrm{Var}(a_y \mid c)$; likewise $b_y = \Pr(Y = y \mid \phi, Z)$ satisfies $\mathrm{Var}(b_y \mid c) \leq \mathrm{Var}(\mathbb{1}\{Y = y\} \mid c) = p_y(c)(1 - p_y(c))$ by the same argument applied to $\mathbb{1}\{Y = y\}$ in place of $H_X(y)$. Both bounds hold termwise, so summing over y and applying the (monotone) square root to each sum separately gives the observable, per-cell bound

$$\sqrt{\textstyle\sum_y \mathrm{Var}(a_y \mid c)}\, \sqrt{\textstyle\sum_y \mathrm{Var}(b_y \mid c)} \leq \sqrt{\textstyle\sum_y \mathrm{Var}(H_X(y) \mid c)}\, \sqrt{\textstyle\sum_y p_y(c)(1 - p_y(c))},$$

the second inequality of Eq. (21); this is a bound on a product of two sums, and does not simplify to a sum of per-label square roots, since Cauchy–Schwarz runs the other way for that rearrangement. Second, for Corollary 2, $a_y \in [-M, M]$ as a conditional expectation of a $[-M, M]$-valued variable, so Popoviciu's inequality gives $\mathrm{Var}(a_y \mid c) \leq M^2$ for every y and hence $\sum_y \mathrm{Var}(a_y \mid c) \leq KM^2$. For the

b_y the same argument would give $\mathrm{Var}(b_y \mid c) \leq 1/4$ and a total of $K/4$, but that treats the b_y as K unconstrained quantities in $[0, 1]$ when in fact $\sum_y b_y = 1$ almost surely. Using that constraint, $\sum_y b_y^2 \leq \left(\sum_y b_y\right)^2 = 1$ pointwise, while $\sum_y (\mathbb{E}b_y)^2 \geq K^{-1}\left(\sum_y \mathbb{E}b_y\right)^2 = 1/K$ by Cauchy–Schwarz, so

$$\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c) = \mathbb{E}\Big[\sum_y b_y^2\Big] - \sum_y (\mathbb{E}b_y)^2 \leq 1 - \frac{1}{K},$$

which is Eq. (22) and is strictly smaller than $K/4$ for every $K > 2$. Therefore

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \leq |\rho| \sqrt{KM^2}\,\sqrt{1 - 1/K} = |\rho|\, M \sqrt{K-1}.$$

By Eq. (19), the left side of each display is exactly $|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]|$, giving Eq. (21) and Eq. (23). □

B. Influence function: estimand and derivation

The estimand under singleton exclusion.. Cells with $n_c = 1$ identify no within-cell counterfactual and are excluded, so the dataset-level statistic targets the *identified-subpopulation* quantity: writing π_c for the cell probabilities, the target of $\widehat{\Delta R}$ is the π-weighted mean $\theta = \sum_c \pi_c\, \Delta R_c / \sum_c \pi_c$ taken over cells that appear with at least two examples. At finite N the identified set is random; we treat the interval as conditional on identification and report the identified fraction alongside every estimate (99.0% of relations on VG150, 100% on CIFAR-100-LT), so the gap between the conditional and full-population targets is bounded by the reported coverage.

Derivation of Eq. (32).. Fix a cell c and let $\theta_c = \Delta R_c$ with $g_c(z) = \mathbb{E}[A(z, Z') \mid Z' \in c]$ the kernel projection. The classical Hájek projection of the order-two U-statistic $\widehat{\Delta R}_c$ is

$$\widehat{\Delta R}_c = \theta_c + \frac{2}{n_c} \sum_{i \in c} (g_c(Z_i) - \theta_c) + O_p(n_c^{-1}).$$

The dataset statistic weights cells by their example counts, $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \widehat{\Delta R}_c$, and cell membership is itself random (multinomial over cells), so an example i contributes both through the within-cell projection and through the between-cell composition term $\theta_{C_i} - \theta$. Combining the two sources,

$$\widehat{\Delta R} - \theta = \frac{1}{N_*} \sum_i \Big[2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta) \Big] + o_p(N_*^{-1/2}),$$

where the dataset-level remainder requires one condition beyond the per-cell expansion, because C per-cell remainders do not aggregate for free. Writing $C = \#\{c : n_c \geq 2\}$ for the number of eligible cells, the weighted sum of the per-cell degenerate terms is $N_*^{-1} \sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C/N_*)$, which is $o_p(N_*^{-1/2})$ provided

$$C = o(\sqrt{N_*}). \tag{B.1}$$

This is what makes the second-order rate $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$ worth stating: the sharper per-cell rate is what allows a condition on the cell *count* alone, with no further requirement on individual cell sizes. On VG150, $C = 6{,}346$ against $N_* = 227{,}337$, so $C/\sqrt{N_*} \approx 13.3$ — the condition is a statement about the limit, and at these finite sample sizes it is the coverage simulation of Section 5.1, not Eq. (B.1), that supports the reported intervals. Estimating $g_{C_i}(Z_i)$ by the pair-kernel mean $\bar{A}_i$, θ_c by $\widehat{\Delta R}_c$, and θ by $\widehat{\Delta R}$ gives the plug-in influence contribution $\widehat{\psi}_i = 2(\bar{A}_i - \widehat{\Delta R}_c) + (\widehat{\Delta R}_c - \widehat{\Delta R}) = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}$, which is Eq. (32) and has exactly zero sample mean by construction. Its empirical 95% coverage in a VG-like regime (image-clustered examples, heavy-tailed cell sizes with a large mass of near-minimum cells) is validated in Section 5.1.

Proof of Theorem 4.. The linearization above gives $\widehat{\Delta R} - \theta = N_*^{-1} \sum_i \psi_i + o_p(N_*^{-1/2})$, under Eq. (B.1), for the population influence contribution $\psi_i = 2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta)$, which has zero mean by construction and satisfies $|\psi_i| \leq C(M)$ for an absolute constant $C(M)$ depending only on the score bound M in condition (i), since g_{C_i}, θ_{C_i}, and θ are themselves averages of the bounded kernel $A_{ij} \in [-2M, 2M]$. Because clusters are mutually independent (condition (ii)), $T_g = \sum_{i\in g} \psi_i$ for $g = 1, \ldots, G$ are independent, mean-zero random variables with $|T_g| \leq |g|\, C(M)$. Condition (ii)'s finite third moment for cluster sizes and $\max_g |g| = o(\sqrt{G})$ give $\sum_g E|T_g|^3 \leq C(M)^3 \sum_g \mathbb{E}|g|^3 = O(G)$ while $(\sum_g \mathrm{Var}(T_g))^{3/2} = \Theta(G^{3/2})$ under condition (iv), so Lyapunov's ratio $\sum_g E|T_g|^3/(\sum_g \mathrm{Var}(T_g))^{3/2} \to 0$; the Lyapunov condition implies the Lindeberg condition, and the Lindeberg–Feller central limit theorem for triangular arrays of independent summands [63, 64] gives $(\sum_g T_g)/\sqrt{\sum_g \mathrm{Var}(T_g)} \xrightarrow{d} \mathcal{N}(0, 1)$. Condition (iii) makes $N_*/\sum_g \mathrm{Var}(T_g) \to 1/\sigma^2$ in probability by definition of σ^2, and the $o_p(N_*^{-1/2})$ remainder is asymptotically negligible after multiplying by $\sqrt{N_*}$, so $\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ by Slutsky's theorem. For the variance estimator, $\widehat{\Delta R} \to_p \theta$ by the weak law of large numbers across the diverging number of identified examples, and $\widehat{\Delta R}_c \to_p \Delta R_c$ for each eligible cell by the weak law of large numbers applied within that cell – this is where condition (iv) is used, and it is not implied by anything aggregate: (iv) guarantees $n_c \to \infty$ almost surely for every cell in the fixed, finite support of ϕ, which is what a within-cell WLLN requires, whereas a condition on the total identified count bounds no individual cell's size. So $\widehat{\psi}_i \to_p \psi_i$ pointwise, and $\widehat{\sigma}^2$ – a sum of squares of cluster-aggregated $\widehat{\psi}_i$, normalized by N_* and G – is a continuous functional of these consistent plug-ins, hence $\widehat{\sigma}^2 \to_p \sigma^2$ by the continuous mapping theorem. □

Efficient computation.. The implementation does not instantiate all pairs. In addition to Eq. (30), let $G_c(y) = \sum_{j\in c} H_j(y)$ and $O_c = \sum_{j\in c} H_j(y_j)$. The pair-kernel mean for observation i is

$$\bar{A}_i = \frac{1}{2}\left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1}\right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters g, the implemented variance is

$$\widehat{\mathrm{Var}}(\widehat{\Delta R}) = \frac{G}{G-1}\frac{1}{N_*^2}\sum_{g=1}^{G}\left(\sum_{i\in g}\widehat{\psi}_i\right)^2,$$

where G is the number of images and $\widehat{\psi}_i$ is Eq. (32). We use a normal critical value; with 27,955 image clusters, small-G corrections are immaterial.

C. Randomization algorithm

For each cell, sort examples once and index them $0, \ldots, n_c - 1$. Randomization b draws $o_c^{(b)}$ uniformly from this index set and maps label position r to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (30) is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero p-values.

D. Implementation and reproducibility

The NumPy implementation is in `wager/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/covariance values plus intervals. The Visual Genome driver is `experiments/run_sgg_wager.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`. All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 4 and 2 (the latter against an explicit discrete population with exact expectations, independent of the estimator's own code path), Corollary 1's log-score covariance identity, Corollary 3's equivalence to the (unclustered) Diebold–Mariano statistic at a trivial grouping variable, and Proposition 3's bound ordering (exact bias $\leq$ tight, per-cell bound $\leq$ crude, data-free bound) on a second explicit discrete population with a known omitted confounder. The covariance-identity cross-check quoted in Appendix 5.7 is reproduced by `experiments/cifar_covariance_check.py`; the calibration-matched protocol and the prior-matching experiment are `experiments/cifar_recalibration_control.py` and `experiments/vg_prior_consequence.py`. Table 3's free-bound rows are reproduced by `experiments/sensitivity_bound_check.py` and its sharp rows, including the robustness value $\rho^{\dagger}$ of Eq. (24), by `experiments/sensitivity_analysis.py`; the evaluation of Corollary 2 is `experiments/sensitivity_bound_check.py`, which recomputes the crude bound directly from the committed `antisymmetric_ablations.json` rather than rerunning the decomposition; `experiments/sensitivity_analysis.py` computes Proposition 3's tighter, per-cell form from per-instance arrays (Section D.1).

Full-data VG predictors (Appendices 5.3–5.5).. FREQ uses add-one smoothing over the 50 predicates within each training class pair, backing off to the subject-marginal and then the global predicate histogram for pairs unseen in training. The MLPs consume fixed random class embeddings (32 dimensions per class, seed 0, concatenated for subject and object) on standardized features and train with AdamW (learning rate 10^{-3}, weight decay 10^{-5}, batch size 4096): MLP-CLASS with hidden sizes (256, 128) for 12 epochs, MLP-SPATIAL with (256, 128) for 15 epochs, and MLP-SPATIAL+ with (512, 256, 128) for 20 epochs – the "+" variant differs in both width and schedule, as noted in Appendix 5.3.

Data availability.. The repository ships the estimator, every driver script, and the cached results files that every printed number traces to (`results/*.json`). The cached per-relation probability arrays that the drivers consume (`sgg_dists.npz`, `vg_visual_models.npz`, `cifar_lt_results.npz`) are regenerable from the committed training scripts with fixed seeds and are archived alongside the code release, so exact reproduction does not require retraining.

D.1. Compute environment for Appendices 5.6–5.7

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The WAGER decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Appendix 5.3. All Python code runs under a pinned environment (Python 3.13).

Matched-subsample models (Appendix 5.6).. The three compared predictors share a fixed, seeded (seed = 0) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate 10^{-3}, weight decay 10^{-5}, 15 epochs, batch size 4096, hidden sizes (256, 128), on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to 224×224 with CLIP's published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as `ViT-B-32-quickgelu`; loading the same weights under the plain `ViT-B-32` configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered afterwards: the 659,210 crops implied by the audited relations collapse to 422,143 unique ones, removing 36.0% of the encoder work without changing any feature. Only the images referenced by those relations are retrieved, individually rather than through the two bulk archives; of the 71,990 required, three were unavailable upstream and fall back to a blank crop.

ResNet-32 triple (Appendix 5.7).. CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. [56] at imbalance ratio 100: class c (zero-indexed, $0, \dots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500 images and class 99 keeps 5; the balanced 10,000-image test split is untouched. All three models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay 2×10^{-4}), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the runs is the per-example loss weight. CE uses uniform weights throughout. CB applies the effective-number class-balanced weight $w_c \propto (1 - \beta)/(1 - \beta^{n_c})$ with $\beta = 0.9999$, renormalized to average to one across classes, from initialization. DRW trains with uniform weights until epoch 60 and switches to exactly the CB weights for the final twenty epochs [57]; no other hyperparameter changes at the switch. The robustness matrix of Table 10 repeats this recipe verbatim at seeds 1 and 2 (ratio 100) and at ratios 10 and 50 (seed 0), re-drawing the long-tailed subset with the configuration's own seed and initializing all three arms of a configuration identically (`experiments/colab_cifar_multiseed.py`). The post-hoc LA arm is not trained: it rescales the baseline's test probabilities by the inverse training class prior, $q_{\mathrm{LA}}(y \mid x) \propto q_{\mathrm{CE}}(y \mid x)/\pi_{\mathrm{train}}(y)$, renormalized.